Devin AI Export - PDF Version
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Overview

Relevant source files

- [

  README.md](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/README.md (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/README.md))

- [

  package.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json))

- [

  src/config.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts))

- [

  src/index.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts))

Purpose and Scope

This document provides a high-level overview of the `lotte-ws-bridge` system, explaining its role as a WebSocket bridge service for financial trading operations. The overview covers the system's primary purpose, key capabilities, architecture, and technology stack.

For detailed setup and deployment instructions, see Getting Started. For in-depth architectural details, see System Architecture. For configuration management details, see Configuration Management.

System Purpose

The `lotte-ws-bridge` is a Node.js service that acts as a bridge between WebSocket clients and various backend financial services. It facilitates real-time communication for trading operations while providing comprehensive notification capabilities for financial events.

The system serves as a central integration point that:

- Maintains persistent WebSocket connections to external trading services
- Processes and routes financial event notifications through Kafka
- Manages real-time data flow between multiple backend systems
- Provides template-based notification rendering for various financial events

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L2-L4)

package.json2-4 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L2-L4)[

src/config.ts29-31](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)%5B)

src/index.ts1-19](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19))

Key Capabilities

Capability	Description
WebSocket Bridge	Maintains connection to Lotte WebSocket server at `ws://172.33.30.23:9900`
Message Queuing	Integrates with Kafka for reliable message processing and routing
Notification System	Comprehensive notification engine supporting 15+ financial event types
Data Persistence	Connects to MySQL databases via TypeORM for data storage
Caching Layer	Redis integration for high-performance data caching
Object Storage	MinIO integration for file and document management
Scheduled Tasks	Node-schedule integration for time-based operations
Health Monitoring	Built-in health checks and comprehensive logging

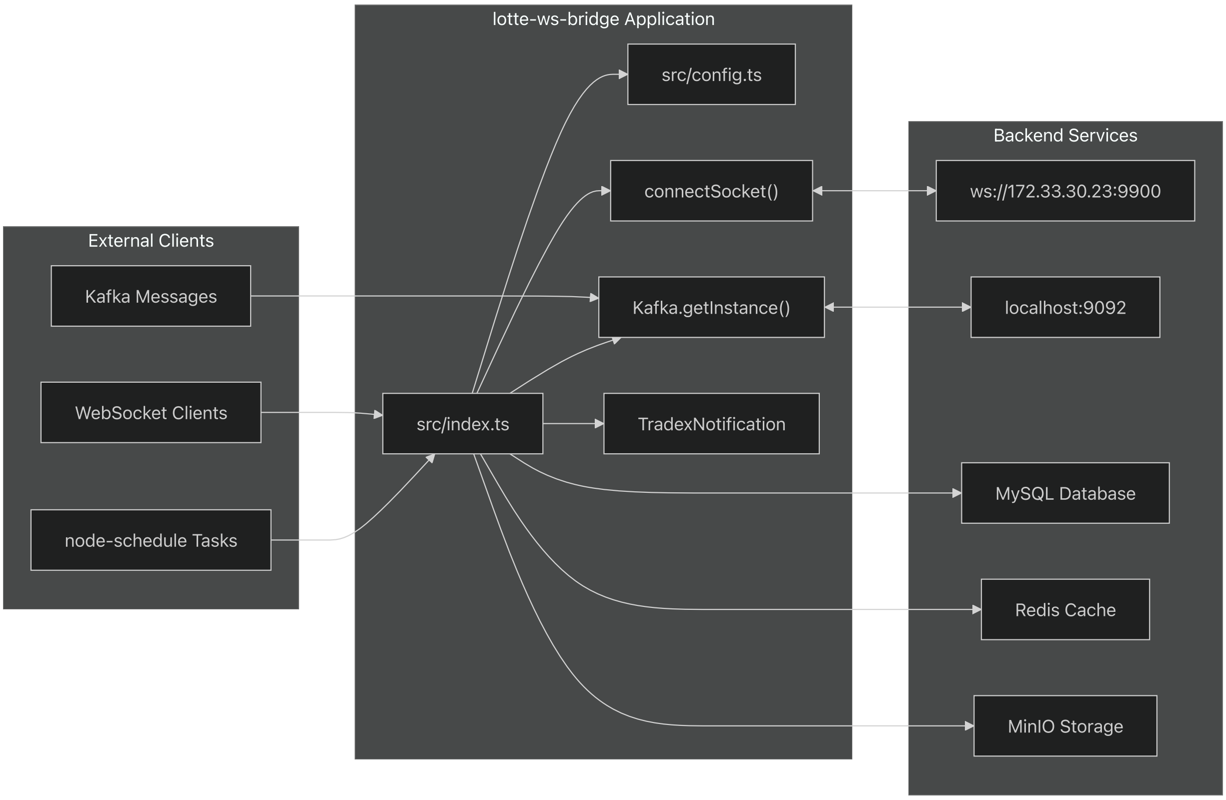
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54) package.json36-54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54)[

src/config.ts10-38](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L38)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L38)%5B)

src/index.ts8-12](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12))

High-Level Architecture

Core System Flow



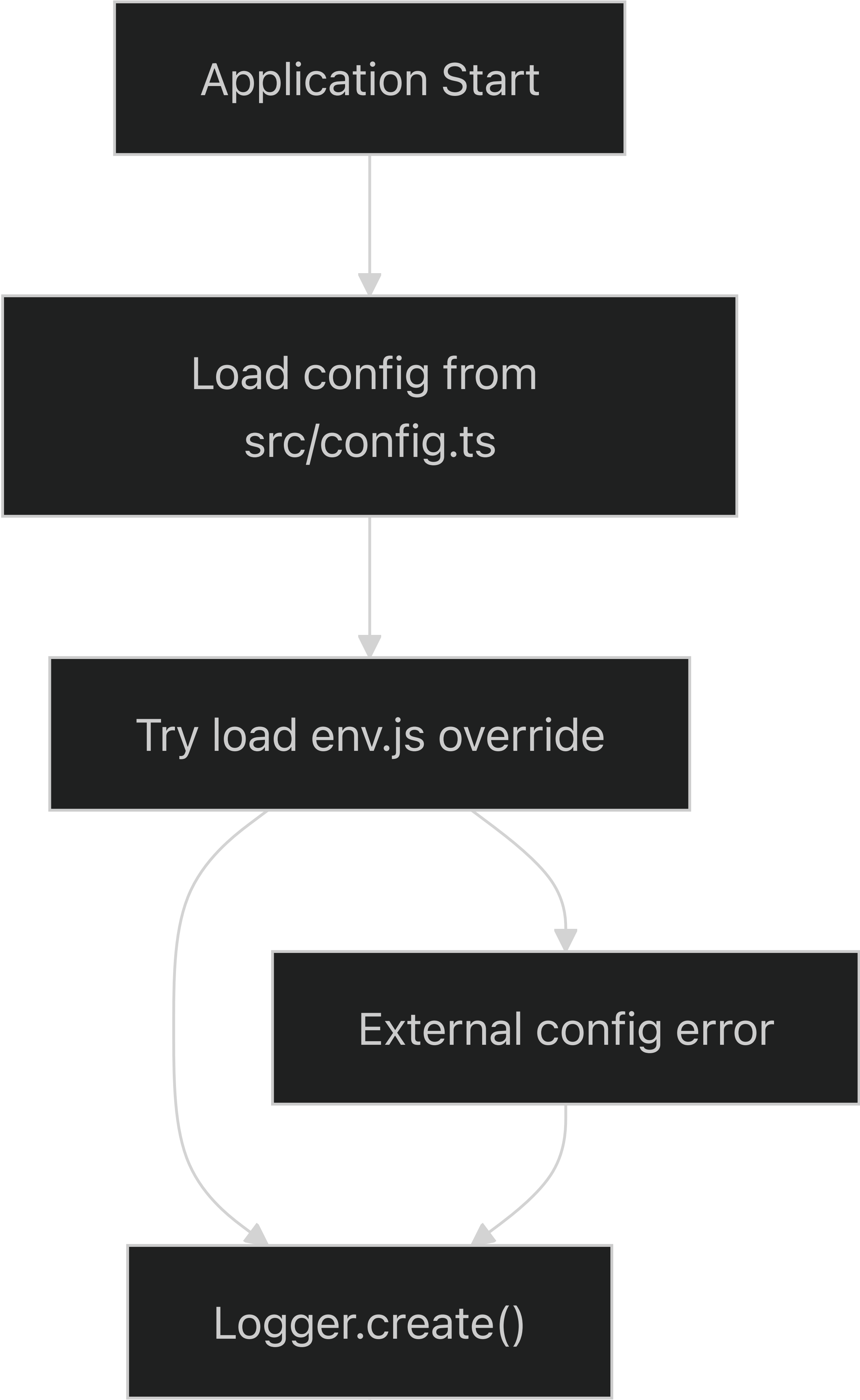
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)

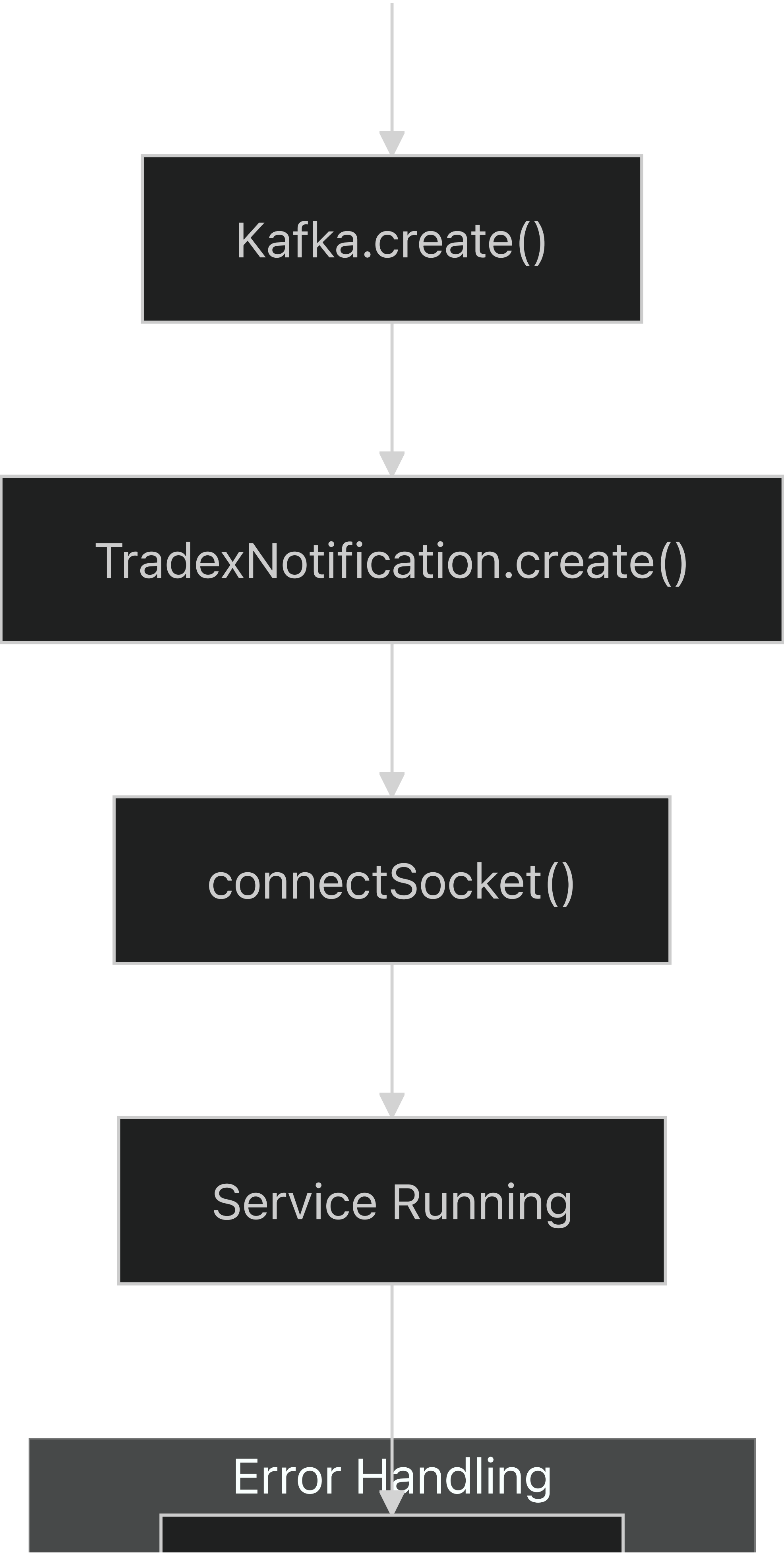
src/index.ts1-19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)[

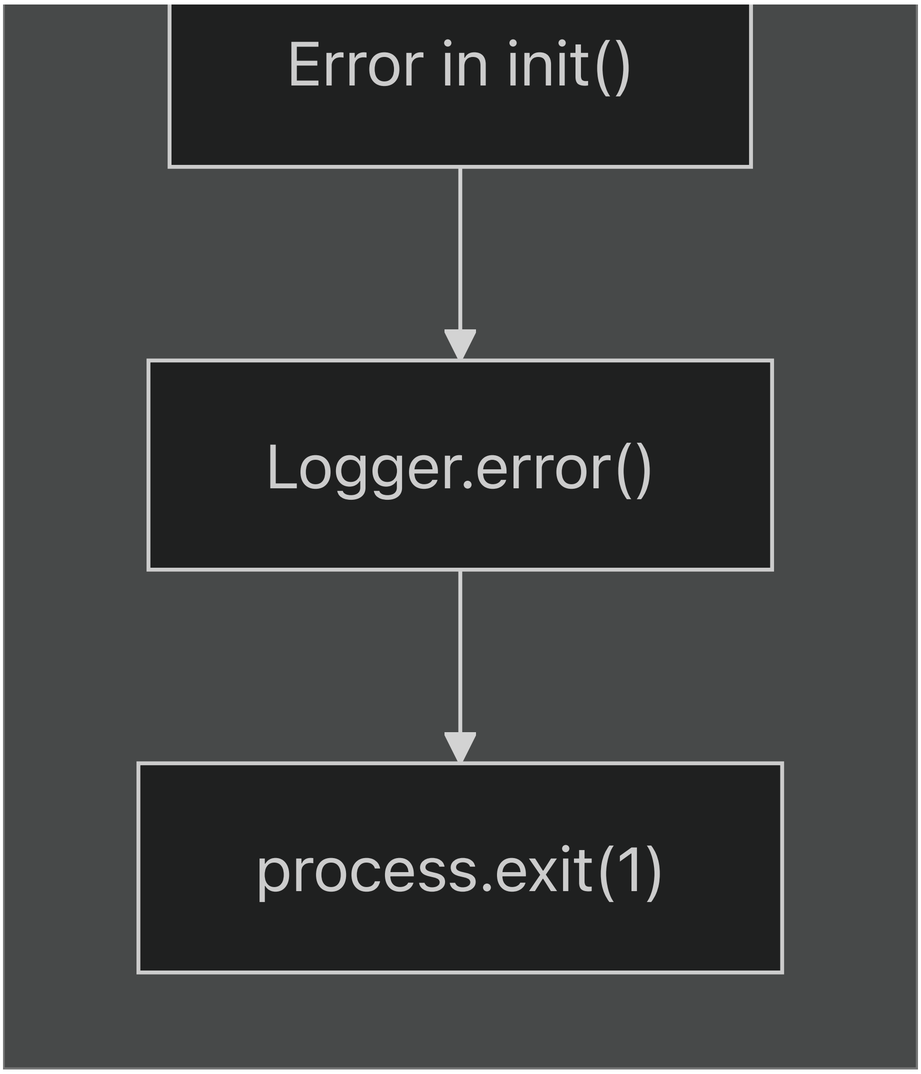
src/config.ts29-31](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)%5B)

package.json36-54](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54))

Application Initialization Flow


Application Start
Load config from src/config.ts
Try load env.js override
External config error
Logger.create()


Kafka.create()
TradexNotification.create()
connectSocket()
Service Running
Error Handling



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L5-L19)

src/index.ts5-19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L5-L19)[

src/config.ts41-53](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53))

Technology Stack

Core Dependencies

Component	Technology	Purpose
Runtime	Node.js 10.16.3	JavaScript runtime environment
Language	TypeScript 4.0.2	Type-safe JavaScript development
WebSocket	ws 8.15.1	WebSocket client implementation
Message Queue	Kafka (via tradex-common)	Event streaming and message processing
Database ORM	TypeORM 0.2.29	Object-relational mapping for MySQL
Caching	Redis 3.0.2	In-memory data structure store
HTTP Client	axios 0.21.4, node-fetch 2.6.0	HTTP request handling
Task Scheduling	node-schedule 1.3.2	Cron-like task scheduling
Object Storage	MinIO 7.0.18	S3-compatible object storage client

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54)

package.json36-54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54)[

.nvmrc](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc)) (from context)

Development Tools

Tool	Version	Purpose
Code Style	Google TypeScript Style (gts) 1.1.2	Code formatting and linting
Linting	TSLint 5.20.1	TypeScript code quality
Testing	Mocha 8.2.0	Unit testing framework
Build	TypeScript Compiler	Compilation to JavaScript
Git Hooks	Husky 1.3.1	Pre-commit code quality checks

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L34)

package.json22-34 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L34)[

package.json5-18](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18))

Core Components

Configuration System

The system uses a hierarchical configuration approach with default settings in `config.ts` and optional environment-specific overrides:

- **Base Configuration**:[

  src/config.ts7-39](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39))

- **Dynamic Override**: External `env.js` file processed via VM context[

  src/config.ts41-53](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53))

- **Kafka Integration**: Merges common, consumer, and producer options[

  src/config.ts60-67](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67))

Service Initialization

The main entry point follows a structured initialization pattern:

1. **Logger Setup**: Creates structured logging with file and console appenders
2. **Kafka Connection**: Establishes connection with configured brokers and options
3. **Notification System**: Initializes template-based notification engine
4. **Socket Connection**: Connects to Lotte WebSocket service

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L5-L12)

src/index.ts5-12 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L5-L12)

Connection Management

The system maintains persistent connections with automatic reconnection and health checking:

- **Ping Cycles**: Configurable ping intervals (`pingCycle: 10`)
- **Duplicate Detection**: Message deduplication (`duplicateCheckSize: 300`)
- **Receive Monitoring**: Connection health checks (`checkReceivedCycle: 3`)
- **Resubscription**: Periodic resubscription cycles (`resubscribeCycle: 900`)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L38)

src/config.ts32-38 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L38)

Getting Started

Relevant source files

- [

  .nvmrc](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc))

- [

  dockerfile](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile))

- [

  entrypoint.sh](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh))

- [

  package.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json))

This document provides step-by-step instructions for setting up, building, and running the lotte-ws-bridge system in both local development and containerized environments. It covers prerequisites, build processes, and basic configuration needed to get the WebSocket bridge application operational.

For detailed system architecture information, see System Architecture. For comprehensive configuration options, see Configuration Management.

Prerequisites

The lotte-ws-bridge application requires specific versions and tools for proper operation:

System Requirements

Component	Version	Source
Node.js	10.16.3	[
.nvmrc1](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1))		
npm	Compatible with Node 10.16.3	[

package.json1-74](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L1-L74 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L1-L74)) | | Docker | Latest (for containerized deployment) | [

dockerfile1-57](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57)) | | TypeScript | 4.0.2+ | [

package.json34](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L34-L34 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L34-L34)) |

Development Tools

The application uses Google TypeScript Style (GTS) and additional linting tools:

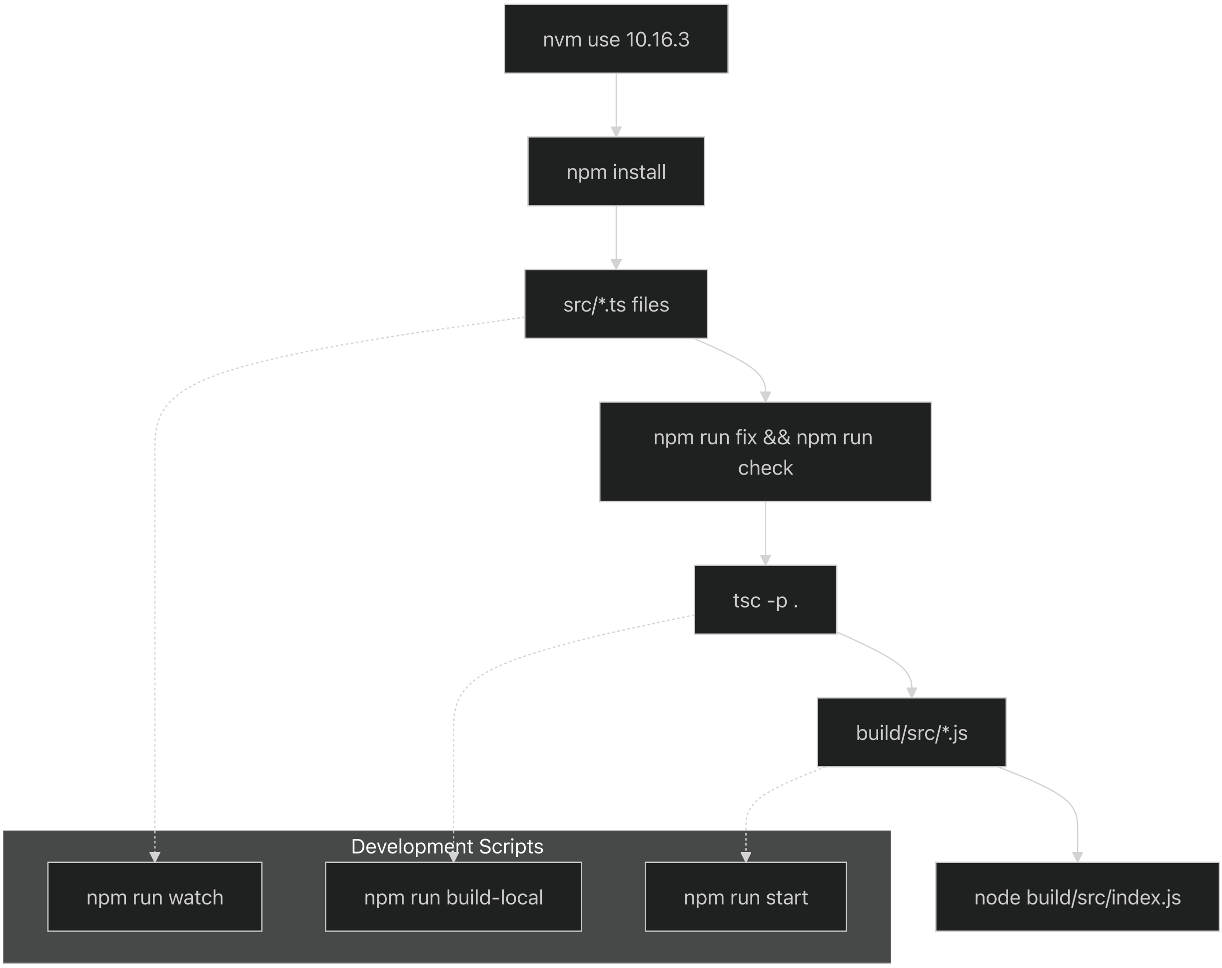
- `gts` - Google TypeScript Style checker and fixer
- `tslint` - TypeScript linting with Microsoft contrib rules
- `prettier` - Code formatting
- `mocha` - Unit testing framework

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35)

package.json22-35 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35)

Local Development Setup

Development Workflow Overview



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18)

package.json5-18 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18)[

.nvmrc1](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1))

Step 1: Node.js Version Setup

Use Node Version Manager to ensure the correct Node.js version:

```
nvm use

nvm use 10.16.3
```

The required version is specified in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)

.nvmrc1 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)

Step 2: Install Dependencies

```
npm install
```

This installs both production and development dependencies defined in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54)

package.json36-54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L54) and[

package.json22-35](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35))

Step 3: Build and Run Locally

The application provides several build and run options:

Command	Purpose	Script Reference
`npm run build`	Full build with linting and compilation	[
package.json7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L7-L7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L7-L7))		
`npm run start`	Build and start application	[

package.json11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L11-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L11-L11)) | | `npm run build-local` | Compile and run from build directory | [

package.json13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L13-L13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L13-L13)) | | `npm run watch` | Watch mode for development | [

package.json8](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L8-L8 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L8-L8)) |

For initial setup, use:

```
npm run start
```

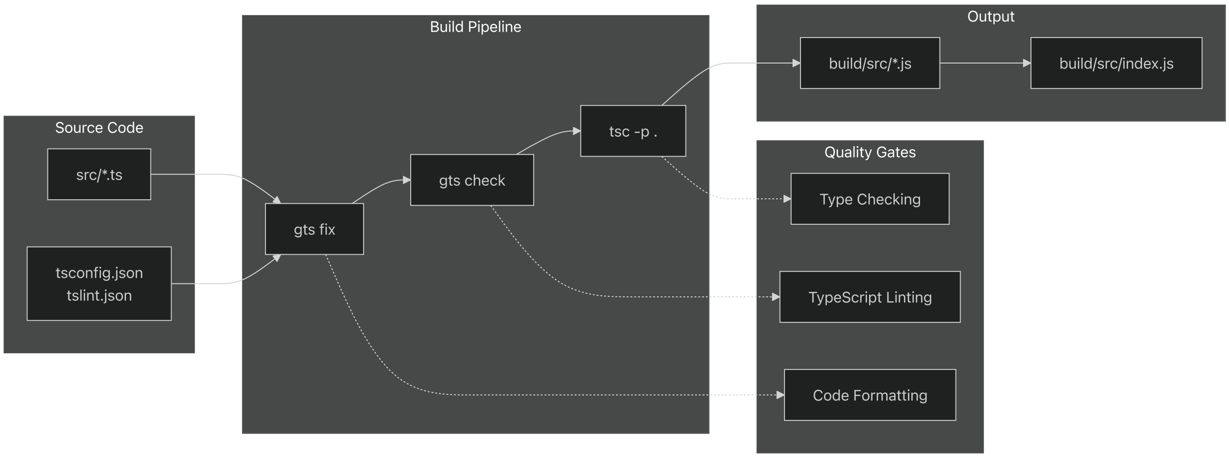
This executes the full build pipeline: `pretest` → `fix` → `check` → `compile` → run.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L10-L13)

package.json10-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L10-L13)

Build Process Details

TypeScript Compilation Flow



The build process enforces code quality through multiple stages as defined in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L10-L10)

package.json10 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L10-L10)

Pre-commit Hooks

Husky manages pre-commit hooks that automatically run formatting and checks:

```
npm run fix && npm run check
```

Configuration in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)

package.json56-60 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)

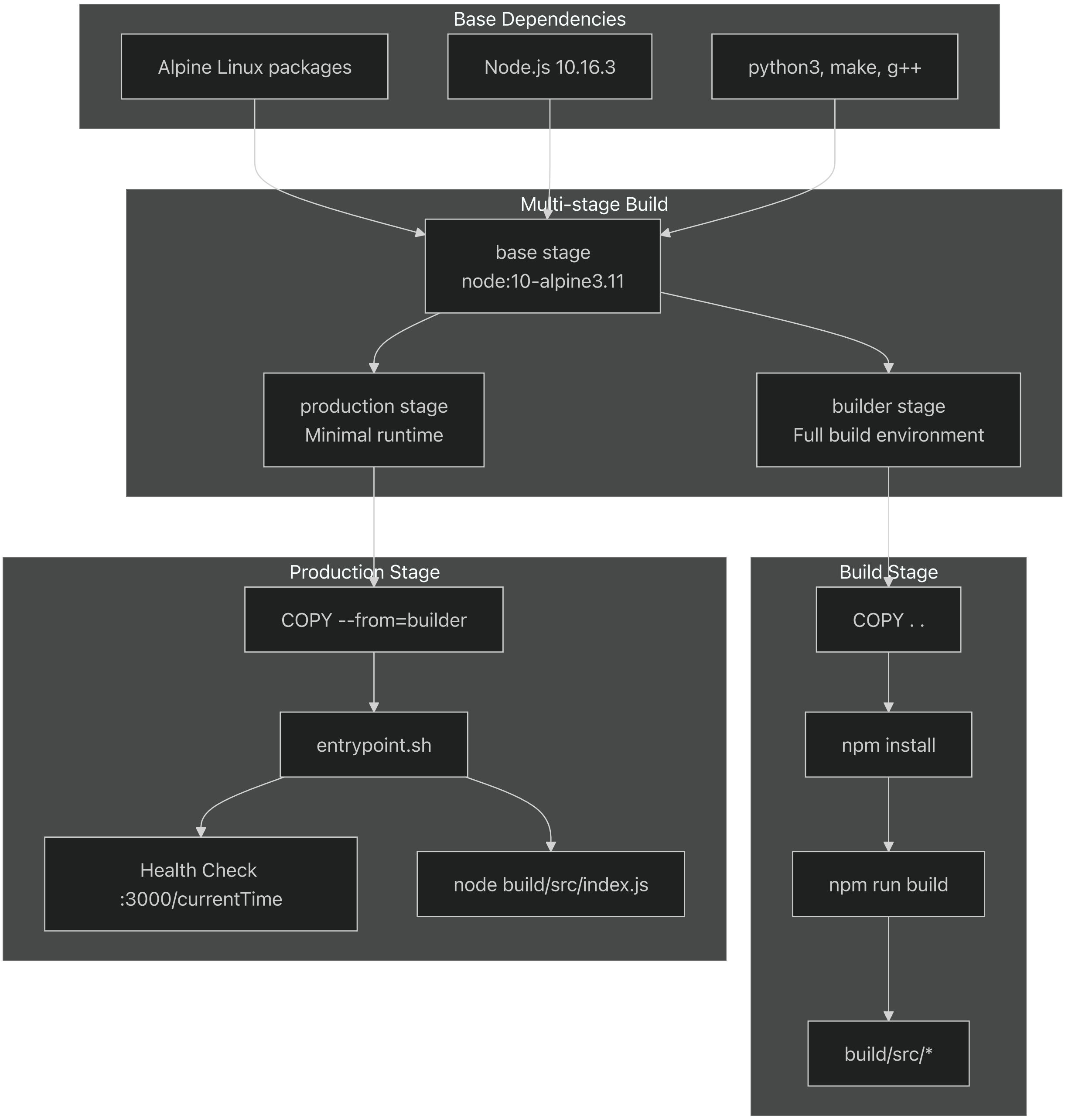
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L6-L18)

package.json6-18 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L6-L18)[

package.json56-60](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60))

Docker Deployment

Container Build Architecture



Building the Docker Image

```
docker build -t lotte-ws-bridge .
```

The Dockerfile uses a multi-stage build pattern defined in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57)

[dockerfile1-57](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57) (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57):

1. **Base stage**: Sets up Alpine Linux with Node.js and build tools[

   dockerfile2-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L2-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L2-L11))

2. **Builder stage**: Installs all dependencies and builds the application[

   dockerfile24-32](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L24-L32 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L24-L32))

3. **Production stage**: Creates minimal runtime image with only built assets[

   dockerfile35-56](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L35-L56 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L35-L56))

Running the Container

```
docker run -p 3000:3000 lotte-ws-bridge
```

The container exposes port 3000 as specified in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L50-L50)

dockerfile50 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L50-L50)

Health Check

The container includes a health check endpoint that verifies the application is responding:

```
curl http://localhost:3000/currentTime
```

Health check configuration in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)

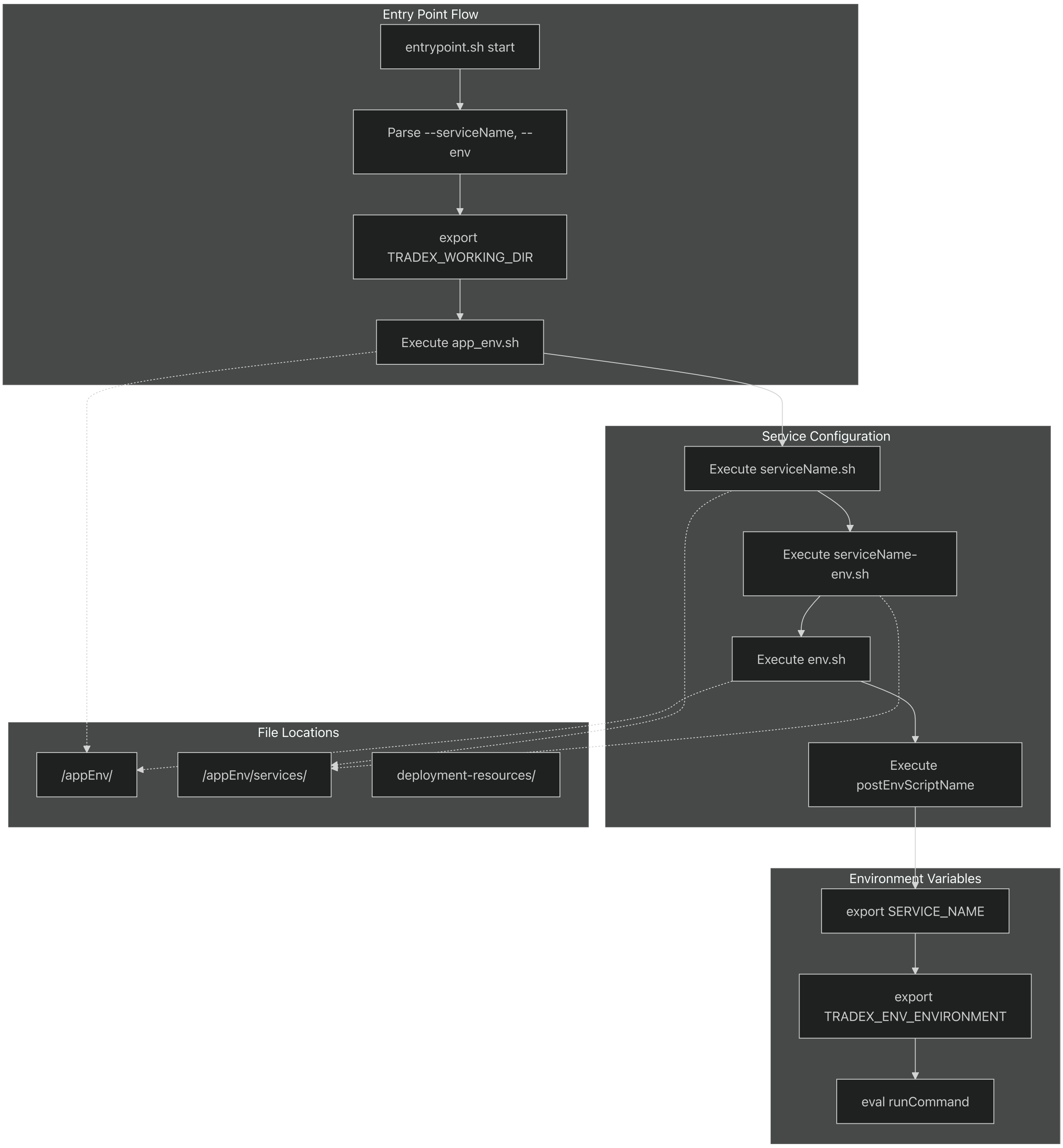
dockerfile53-54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57)

dockerfile1-57 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L57)

Service Orchestration

entrypoint.sh Environment Management



The `entrypoint.sh` script provides sophisticated environment management for different deployment scenarios. Key features:

- Service-specific configuration loading[

  entrypoint.sh44-48](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L44-L48 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L44-L48))

- Environment-specific overrides[

  entrypoint.sh50-54](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L50-L54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L50-L54))

- Resource deployment scripts[

  entrypoint.sh65-74](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L65-L74 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L65-L74))

- Environment variable propagation[

  entrypoint.sh84-87](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L87 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L87))

Example usage:

```
./entrypoint.sh -s lotte-bridge -e production node build/src/index.js
```

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L1-L90)

entrypoint.sh1-90 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L1-L90)

Configuration Overview

Basic Configuration Structure

The application uses a hierarchical configuration system with multiple override layers:

1. **Default configuration** in `src/config.ts`
2. **Environment-specific overrides** via `env.js` VM script execution
3. **Service-specific configuration** through entrypoint.sh orchestration
4. **Runtime environment variables**

For comprehensive configuration details, see Configuration Management.

Key Environment Variables

Variable	Purpose	Set By
`TRADEX_WORKING_DIR`	Application working directory	[
entrypoint.sh35](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L35-L35 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L35-L35))		
`SERVICE_NAME`	Service identifier	[

entrypoint.sh84](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L84 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L84)) | | `TRADEX_ENV_ENVIRONMENT` | Environment identifier | [

entrypoint.sh85-87](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L85-L87 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L85-L87)) |

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L35-L35)

entrypoint.sh35 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L35-L35)[

entrypoint.sh84-87](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L87 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L84-L87))

Verification

Application Startup Verification

After starting the application, verify it's running correctly:

1. **Health Check Endpoint**:

```
```

curl http://localhost:3000/currentTime (http://localhost:3000/currentTime)

```
    Should return current timestamp if application is healthy.

2.  **Log Output**: Check console output for successful initialization messag

3.  **Process Status**:

    ```
# For local development
    ps aux | grep "node.*index.js"

    # For Docker
    docker ps | grep lotte-ws-bridge
```

Common Issues

Issue	Cause	Solution
Node version mismatch	Wrong Node.js version	Use `nvm use 10.16.3`
Build failures	Missing dependencies	Run `npm install`
TypeScript errors	Linting issues	Run `npm run fix`
Port conflicts	Port 3000 in use	Change port or stop conflicting service

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)

[.nvmrc1](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1) (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)[

dockerfile53-54](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)%5B)

package.json14-16](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L14-L16 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L14-L16))

System Architecture

Relevant source files
```

- [

  dockerfile](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile))

- [

  package-lock.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json))

- [

  src/config.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts))

- [

  src/index.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts))

This document describes the architectural design and key components of the `lotte-ws-bridge` application, a Node.js service that acts as a bridge between Lotte WebSocket services and various backend systems including Kafka message queues and notification services.

For information about specific data models used in notifications, see Notification Models. For details about configuration parameters and deployment, see Configuration Management.

Overview

The `lotte-ws-bridge` is a real-time integration service that maintains persistent WebSocket connections to external Lotte services while providing reliable message routing through Kafka and notification processing capabilities. The system operates as a containerized Node.js application with comprehensive logging, health monitoring, and fault tolerance mechanisms.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)

src/index.ts1-19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)[

src/config.ts7-39](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39))

Core System Components

The system follows a layered initialization pattern where the main entry point [`index.ts`] orchestrates the startup of all core components in a specific sequence:

1. **Configuration Loading** - [`config.ts`] loads default settings and applies environment-specific overrides
2. **Logging Initialization** - `Logger.create()` establishes file and console logging
3. **Kafka Setup** - `Kafka.create()` configures message queue connectivity
4. **Notification Engine** - `TradexNotification.create()` enables template-based notifications
5. **WebSocket Connection** - `connectSocket()` establishes the bridge to Lotte services

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12)

src/index.ts8-12 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12)[

src/config.ts18-28](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28))

Data Flow Architecture

The data flow follows a hub-and-spoke pattern where the `SocketClient` serves as the central processing unit. Key processing mechanisms include:

- **Duplicate Detection** - Using `duplicateCheckSize: 300` to track recent messages
- **Health Monitoring** - `pingCycle: 10` second intervals for connection health
- **Message Routing** - Intelligent routing based on message type and content
- **Notification Processing** - Template-based rendering through `TradexNotification`

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L38)

src/config.ts32-38 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L38)[

src/index.ts9-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L11))

Configuration Management Architecture

The configuration system implements a sophisticated override mechanism using VM script execution. The system loads a base configuration and then executes an external `env.js` file in a controlled VM context, allowing dynamic configuration modification without code changes.

Key configuration categories:

- **Kafka Settings** - `kafkaUrls`, consumer/producer options, topic configurations

- **WebSocket Settings** - `websocketAddress`, connection timeouts, retry logic
- **Timing Parameters** - Health check intervals, reconnection cycles, duplicate detection windows
- **Logging Configuration** - File appenders, console output, log rotation settings

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L69)

src/config.ts7-69 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L69)[

src/config.ts41-53](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53))

External System Integration

System	Configuration Key	Purpose	Connection Details
Lotte WebSocket	`websocketAddress`	Primary data source	`ws://172.33.30.23:9900`
Kafka Cluster	`kafkaUrls`	Message queuing	`['localhost:9092']`
File System	`logger.config.appenders.file`	Log persistence	`/logs/application.log`
VM Runtime	N/A	Configuration override	`env.js` script execution

The integration pattern emphasizes fault tolerance and monitoring:

- **Connection Health** - `pingCycle: 10` second WebSocket health checks
- **Retry Logic** - `retryTimes: 3` for failed Kafka operations
- **Timeout Management** - `defaultKafkaTimeout: 10000ms` for message operations
- **Log Rotation** - Automatic rotation with `maxLogSize: 10485760` bytes and `backups: 10`

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L16)

src/config.ts10-16 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L16)[

src/config.ts29-31](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L31)%5B)

src/config.ts18-28](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28))

Deployment Architecture

The deployment follows Docker best practices with multi-stage builds and health monitoring:

- **Build Optimization** - Separate build and production stages for minimal image size
- **Health Monitoring** - HTTP endpoint at `/currentTime` with 30-second intervals
- **Process Management** - `entrypoint.sh` handles service lifecycle and environment setup
- **Port Configuration** - Service exposes port `3000` for health checks and monitoring

The containerized architecture supports both development and production deployments with environment-specific configuration through the VM script system.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L23-L56)

dockerfile23-56 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L23-L56)[

dockerfile52-54](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L52-L54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L52-L54))

Configuration Management

Relevant source files

- [

  entrypoint.sh](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh))
- [

  package.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json))
- [

  src/config.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts))

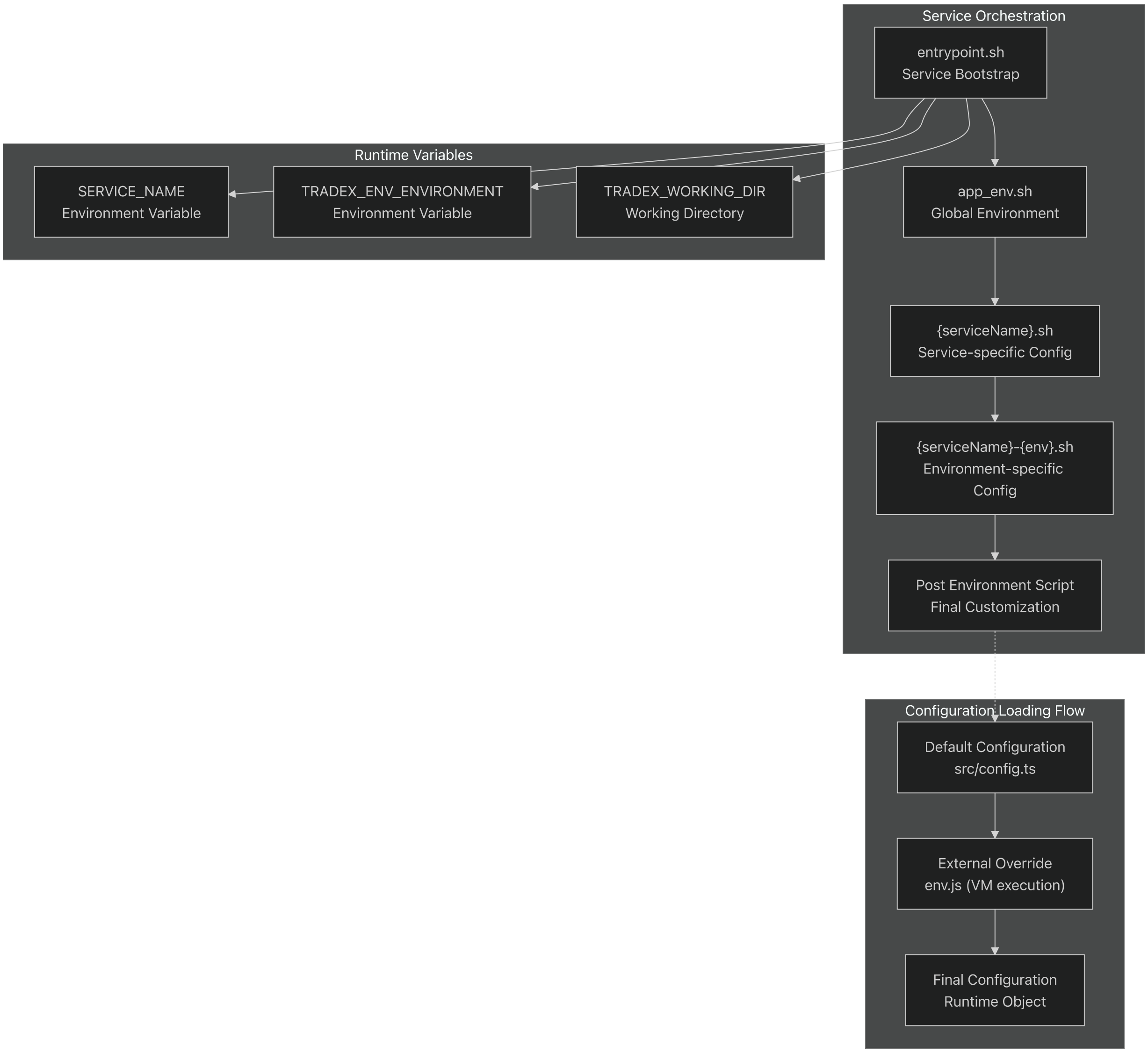
Purpose and Scope

This document covers the comprehensive configuration management system for the `lotte-ws-bridge` application. The system provides a multi-layered configuration approach that combines default settings, external configuration overrides, environment-specific customization, and runtime orchestration.

For information about how these configurations are used in the core application components, see [Core Application Components](). For details about external service integrations that rely on these configurations, see [External Integrations]().

Configuration Architecture

The configuration system operates through multiple layers that are applied in a specific order to create the final runtime configuration:



Sources: [(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L1-L90)](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L1-L90) entrypoint.sh1-90 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L1-L90)[

src/config.ts1-69](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L1-L69 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L1-L69))

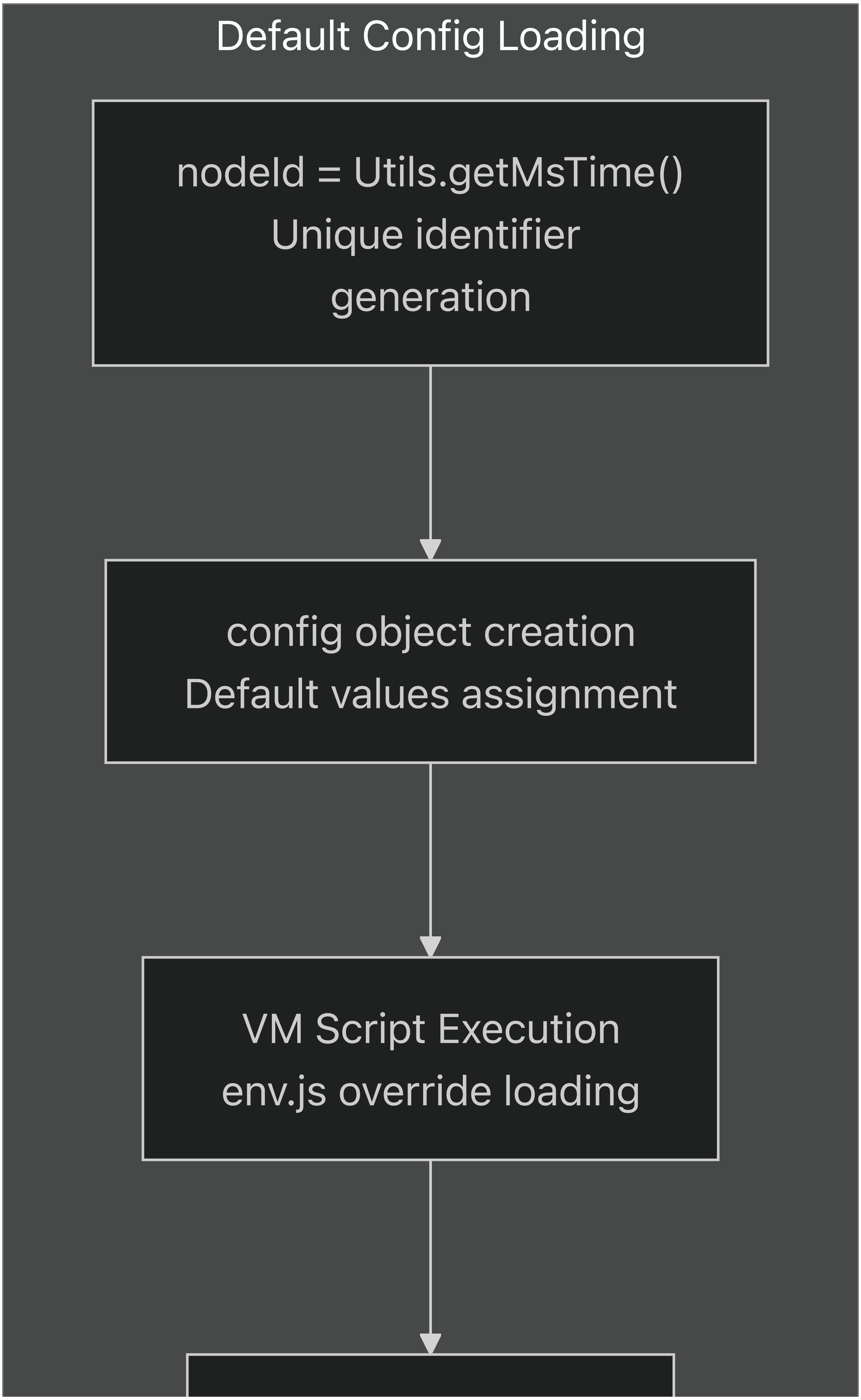
Default Configuration System

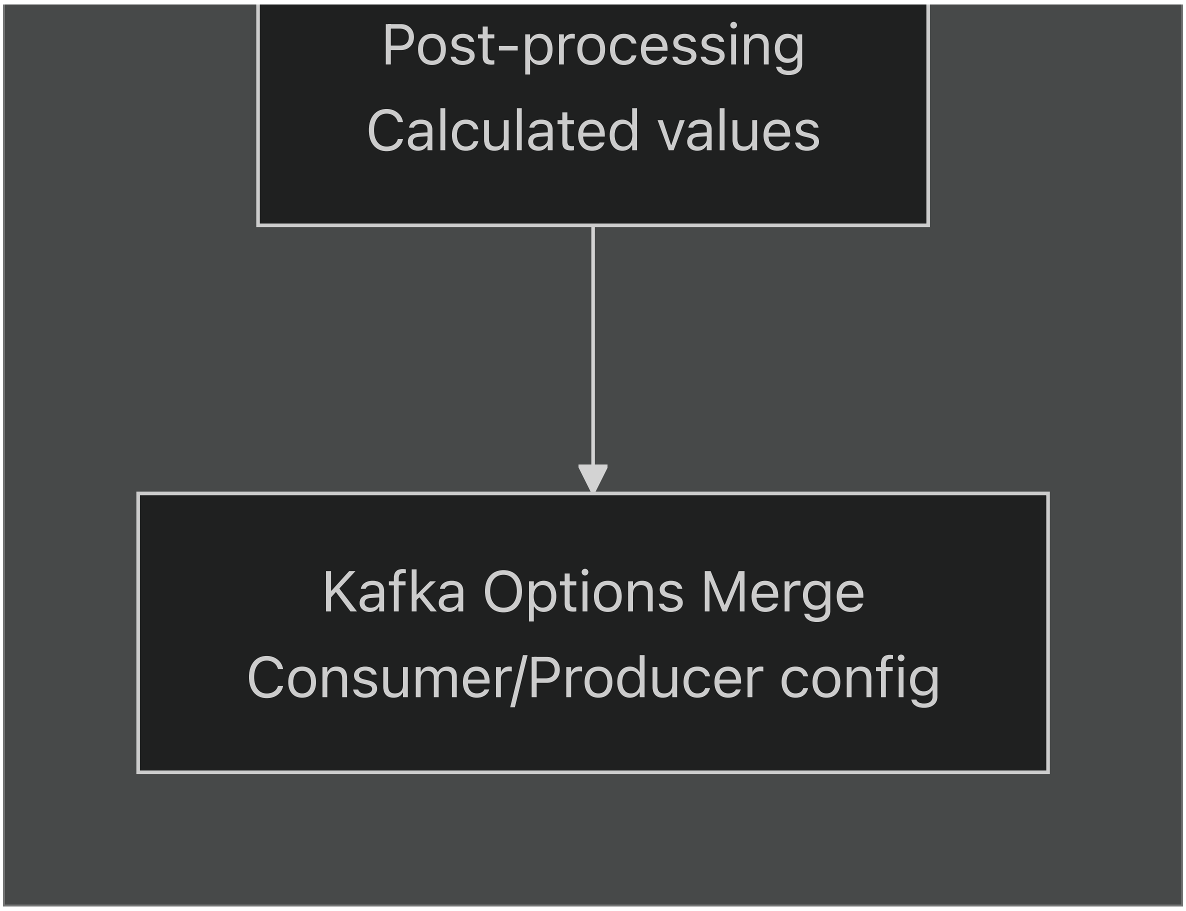
The base configuration is defined in the `config` object within `src/config.ts`. This provides sensible defaults for all system components:

Core Configuration Structure

Configuration Section	Purpose	Key Properties
Cluster Identity	Service identification	`clusterId`, `clientId`
Kafka Configuration	Message broker settings	`kafkaUrls`, `kafkaCommonOptions`, `kafkaConsumerOptions`, `kafkaProducerOptions`
Logging Configuration	Application logging	`logger.config` with appenders and categories
Lotte WebSocket	External service connection	`lotte.websocketAddress`
Timing Parameters	Operational intervals	`pingCycle`, `checkReceivedCycle`, `resubscribeCycle`

Configuration Initialization Process


Default Config Loading
nodeId = Utils.getMsTime()
Unique identifier
generation
config object creation
Default values assignment
VM Script Execution
env.js override loading



The system generates a unique `clientId` using `Utils.getMsTime()` and creates the base configuration object (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L5-L9)

src/config.ts5-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L5-L9) Default values include:

- **Kafka URLs**: `['localhost:9092']` [

  src/config.ts10](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L10 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L10))

- **Lotte WebSocket**: `'ws://172.33.30.23:9900'` [

  src/config.ts30](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L30-L30 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L30-L30))

- **Log File**: `'/logs/application.log'` [

  src/config.ts22](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L22-L22 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L22-L22))

- **Retry Times**: `3` [

  src/config.ts15](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L15-L15 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L15-L15))

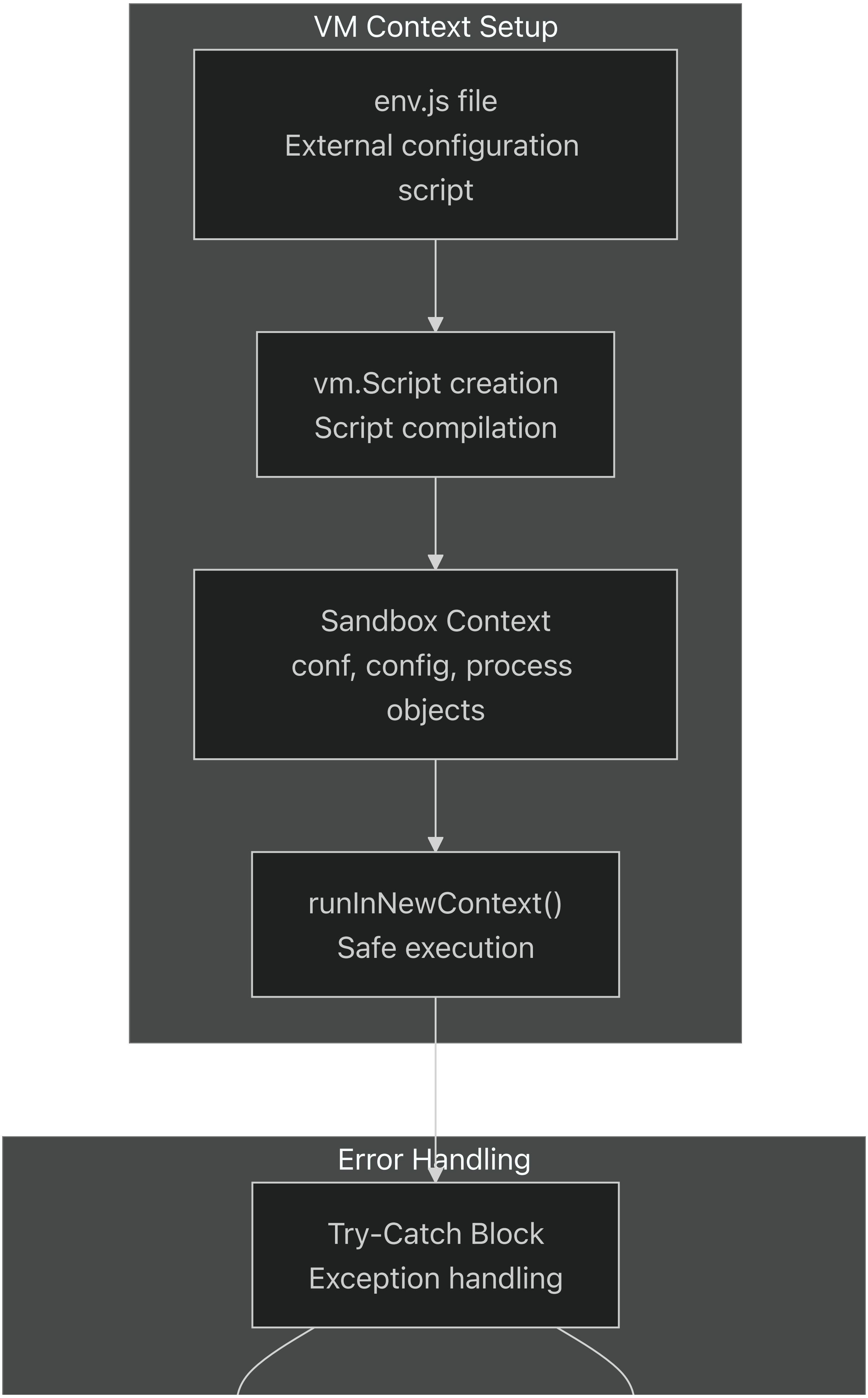
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39)

src/config.ts7-39 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39)

External Configuration Override System

The application supports runtime configuration modification through an external `env.js` file that is executed in a sandboxed VM context:

VM Execution Environment


VM Context Setup
env.js file
External configuration
script
vm.Script creation
Script compilation
Sandbox Context
conf, config, process
objects
runInNewContext()
Safe execution
Error Handling
Try-Catch Block
Exception handling

The VM execution provides access to:

- `conf`: Reference to the configuration object
- `config`: Alias for the configuration object
- `process`: Node.js process object

This approach allows safe external modification of configuration without code changes (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)

src/config.ts41-53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)

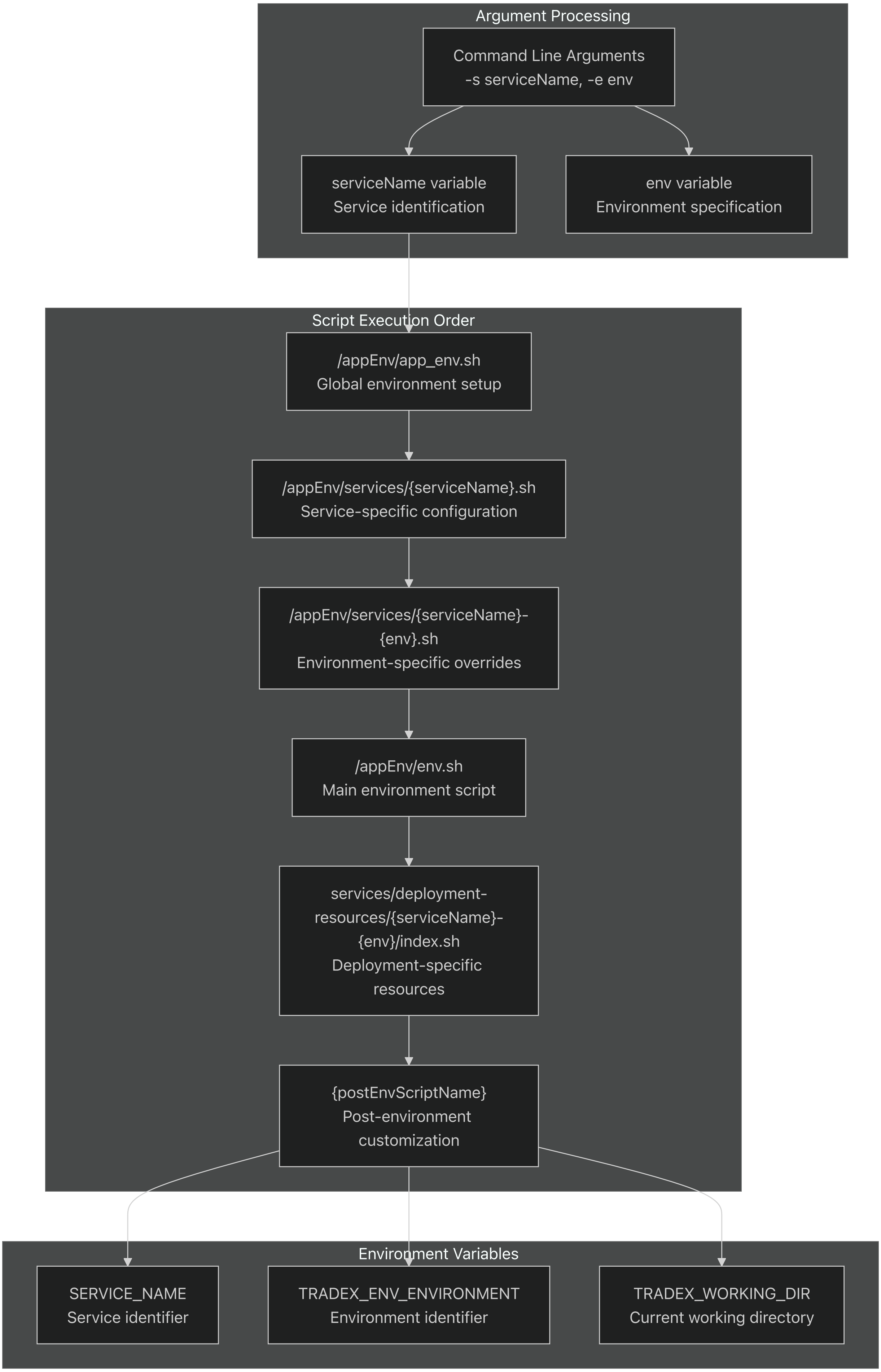
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)

src/config.ts41-53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)

Service Orchestration System

The `entrypoint.sh` script provides sophisticated service orchestration with environment-specific configuration loading:

Script Execution Hierarchy


Argument Processing
Command Line Arguments
-s serviceName, -e env
serviceName variable
Service identification
env variable
Environment specification
Script Execution Order
/appEnv/app_env.sh
Global environment setup
/appEnv/services/{serviceName}.sh
Service-specific configuration
/appEnv/services/{serviceName}-{env}.sh
Environment-specific overrides
/appEnv/env.sh
Main environment script
services/deployment-resources/{serviceName}-{env}/index.sh
Deployment-specific resources
{postEnvScriptName}
Post-environment customization
Environment Variables
SERVICE_NAME
Service identifier
TRADEX_ENV_ENVIRONMENT
Environment identifier
TRADEX_WORKING_DIR
Current working directory

Directory Structure and Script Loading

The orchestration system expects a specific directory structure:

Directory/File	Purpose	Conditional Execution
`/appEnv/app_env.sh`	Global environment setup	Always executed
`/appEnv/services/{serviceName}.sh`	Service-specific configuration	If file exists
`/appEnv/services/{serviceName}-{env}.sh`	Environment-specific overrides	If file exists
`/appEnv/env.sh`	Main environment script	If file exists
`services/deployment-resources/{serviceName}-{env}/`	Deployment resources	If directory exists

The system sets executable permissions and sources each script in sequence (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L37-L82)

entrypoint.sh37-82 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L37-L82)

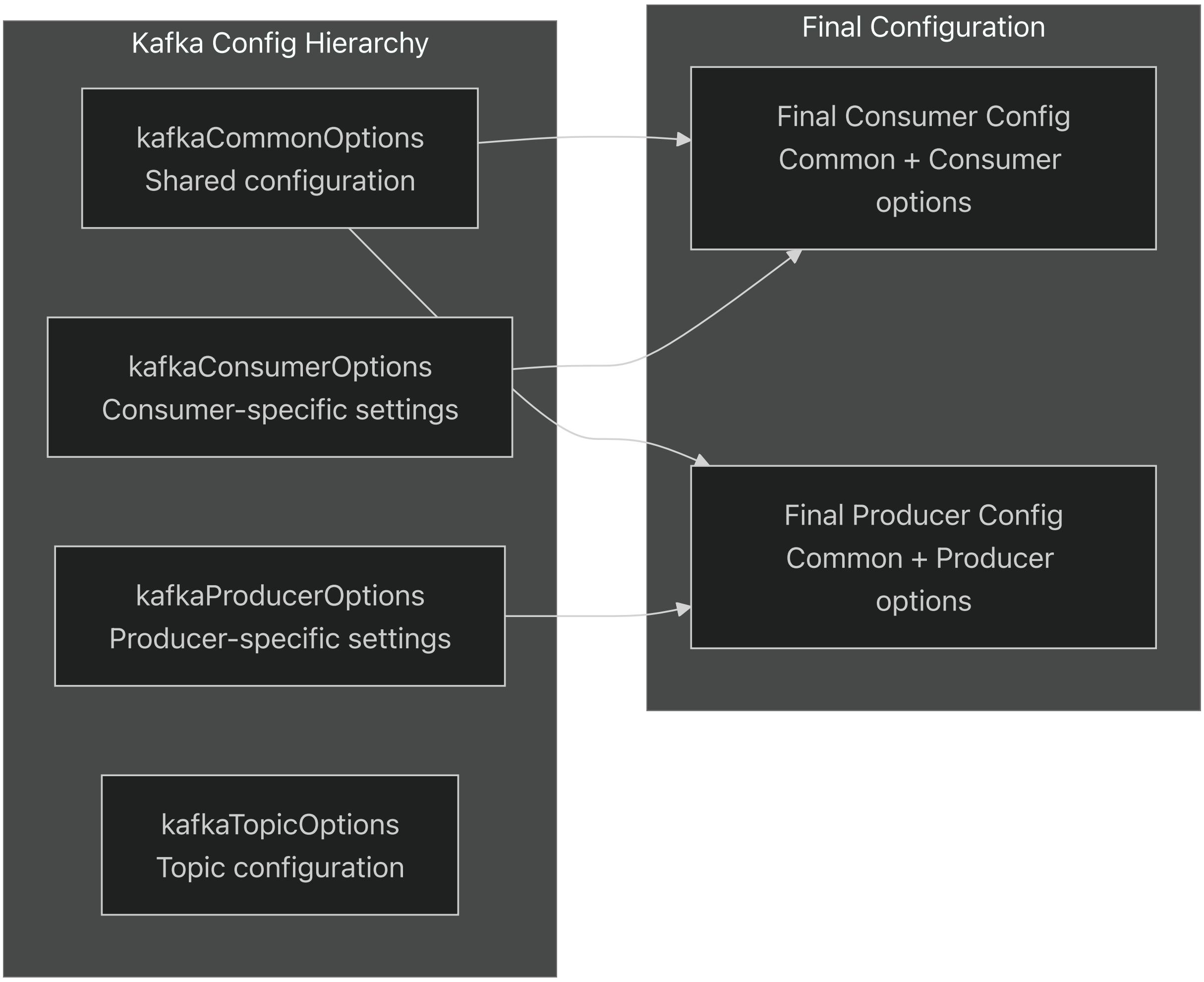
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L7-L90)

entrypoint.sh7-90 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L7-L90)

Configuration Categories

Kafka Configuration

The Kafka configuration system supports hierarchical option merging:



The system merges common options with specific options using spread syntax (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67)

src/config.ts60-67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67)

Logging Configuration

The logging system uses a structured configuration with multiple appenders:

Appender	Type	Configuration
`application`	console	Standard console output
`file`	file	Log file with rotation

File appender settings:

- **Filename**: `/logs/application.log`
- **Compression**: Enabled

- **Max Log Size**: 10,485,760 bytes (10MB)
- **Backups**: 10 files
- **Level**: info

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28)

src/config.ts18-28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28)

Timing and Operational Parameters

The system defines various timing parameters for operational control:

Parameter	Default Value	Purpose
`duplicateCheckSize`	300	Size of duplicate message check buffer
`checkInterval`	1000ms	General check interval
`pingCycle`	10	Ping cycle frequency
`checkReceivedCycle`	3	Received message check cycle
`checkReceivedCycleMs`	6000ms	Calculated from cycle values
`resubscribeCycle`	900	Resubscription interval (0 to disable)

The system performs post-processing calculations to derive additional timing values (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L56)

src/config.ts55-56 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L56)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L39)

src/config.ts32-39 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L32-L39)[

src/config.ts55-56](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L56 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L56))

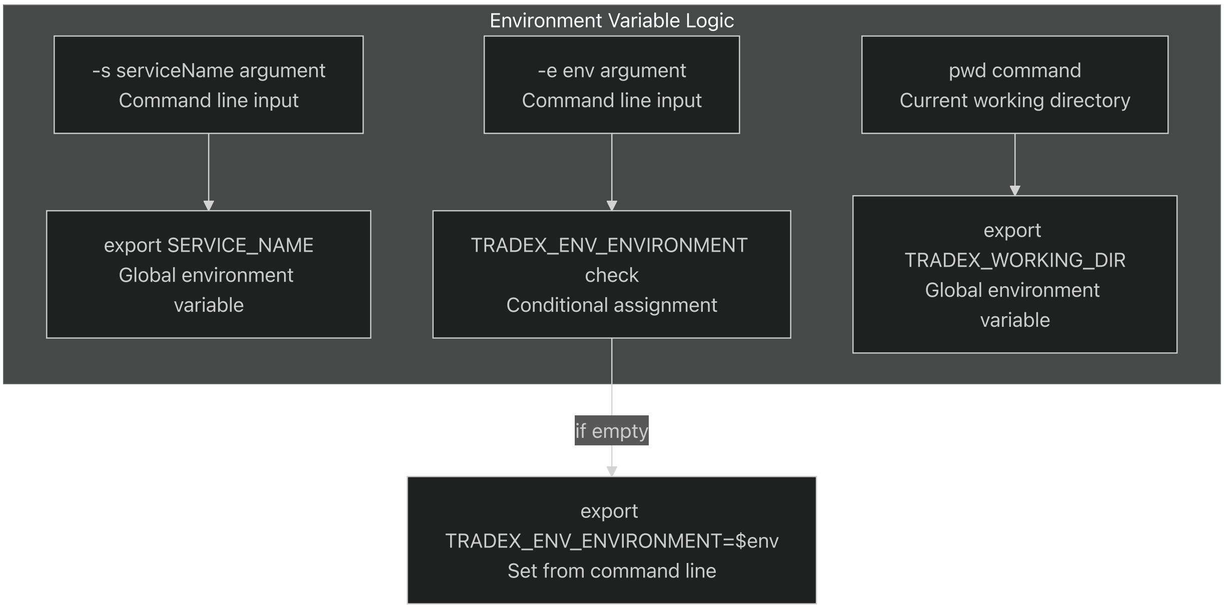
Environment Variables

Core Environment Variables

The orchestration system exports three primary environment variables:

Variable	Source	Purpose
`SERVICE_NAME`	Command line `-s` parameter	Service identification
`TRADEX_ENV_ENVIRONMENT`	Command line `-e` parameter	Environment specification
`TRADEX_WORKING_DIR`	Current working directory	Application root path

Environment Variable Assignment Logic



The system only sets `TRADEX_ENV_ENVIRONMENT` if it's not already defined (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L85-L87)

entrypoint.sh85-87 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L85-L87)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L34-L87)

entrypoint.sh34-87 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/entrypoint.sh#L34-L87)

Configuration Runtime Processing

Post-Processing and Validation

After loading external configurations, the system performs several post-processing steps:

1. **Timing Calculations**: Derives `checkReceivedCycleDouble` and `checkReceivedCycleMs` from base cycle values
2. **Kafka Options Merging**: Combines common options with consumer/producer specific settings
3. **Configuration Logging**: Outputs final configuration for debugging

The system logs the complete configuration after external injection for troubleshooting purposes (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L58-L58)

src/config.ts58 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L58-L58)

Configuration Export

The configuration is exported as the default export, making it available to all application modules:

```
export default config;
```

This allows other modules to import the fully-processed configuration object (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L69-L69)

src/config.ts69 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L69-L69)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L69)

src/config.ts55-69 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L69)

Core Application Components

Relevant source files

- [

  package-lock.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json))
- [

  src/config.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts))
- [

src/index.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts))

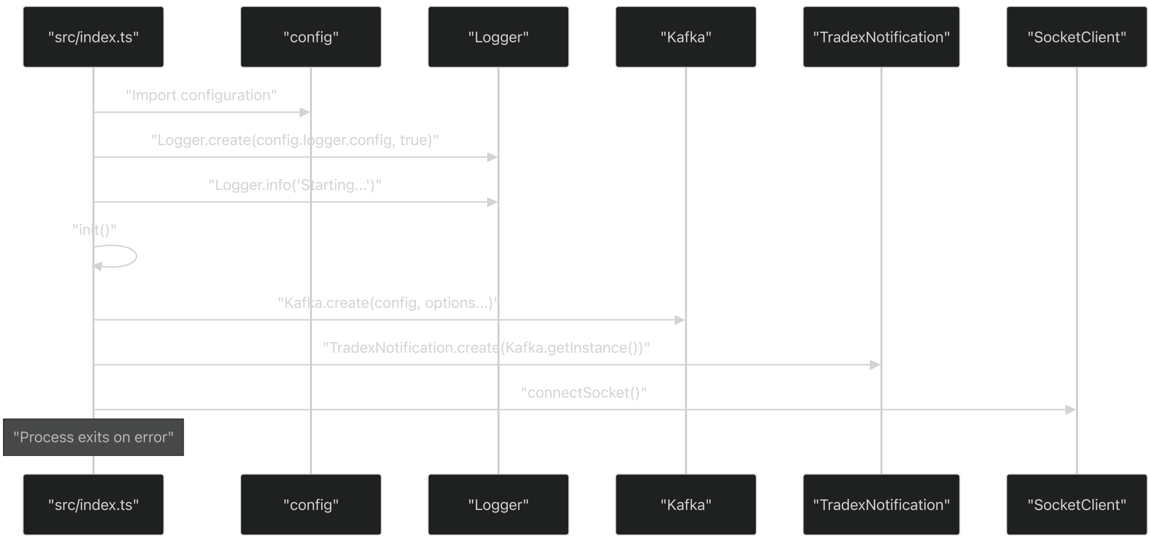
This document details the core components that form the foundation of the lotte-ws-bridge application, including application initialization, configuration management, logging infrastructure, and external service integrations. These components work together to establish the runtime environment and provide essential services for the WebSocket bridge functionality.

For information about the notification system and data models, see Data Models. For details about external service integrations beyond the core setup, see External Integrations.

Application Initialization

The application follows a structured initialization sequence that establishes logging, configuration, and core services before starting the WebSocket bridge functionality.

Initialization Flow



Application Entry Point Flow

The main application entry point in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts%60)

`src/index.ts` (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts%60)() implements a sequential initialization pattern:

Step	Component	Function	Purpose
1	Configuration	Import	Load default and external configuration
2	Logging	`Logger.create()`	Initialize logging infrastructure
3	Kafka	`Kafka.create()`	Establish message queue connection
4	Notifications	`TradexNotification.create()`	Setup notification engine
5	Socket	`connectSocket()`	Start WebSocket bridge

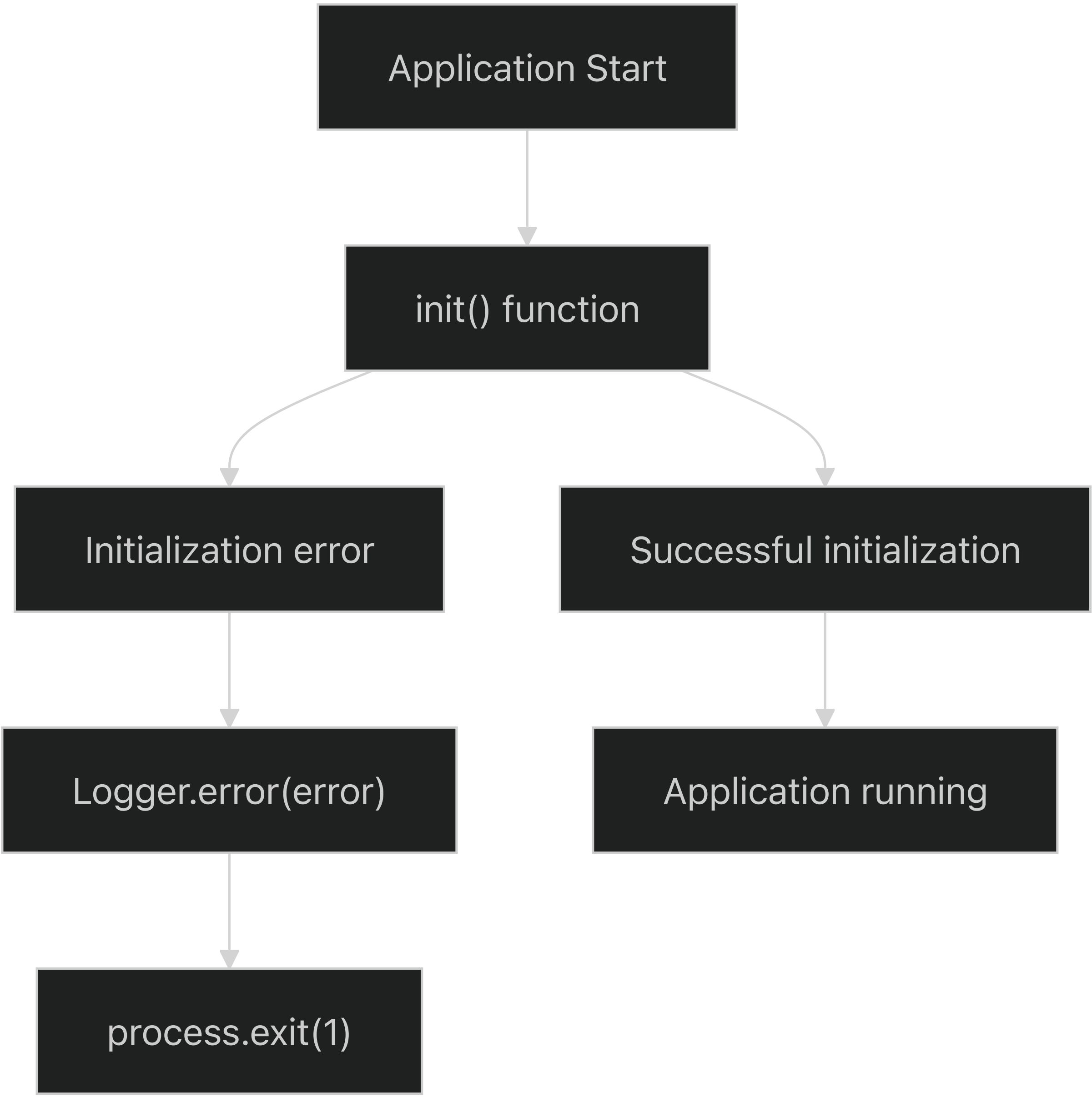
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L1-L20)

`src/index.ts1-20 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L1-L20)[

`src/config.ts1-70](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/`src/config.ts#L1-L70 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L1-L70))

Error Handling and Process Management

The application implements robust error handling at the initialization level:



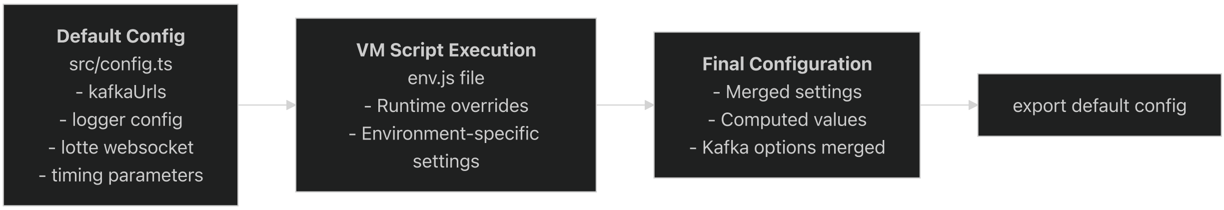
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L14-L19)

` src/index.ts14-19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L14-L19)

Configuration Management

The configuration system provides a hierarchical approach to application settings, supporting both default values and runtime overrides through external scripts.

Configuration Architecture



Core Configuration Sections

The default configuration object in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L7-L39)

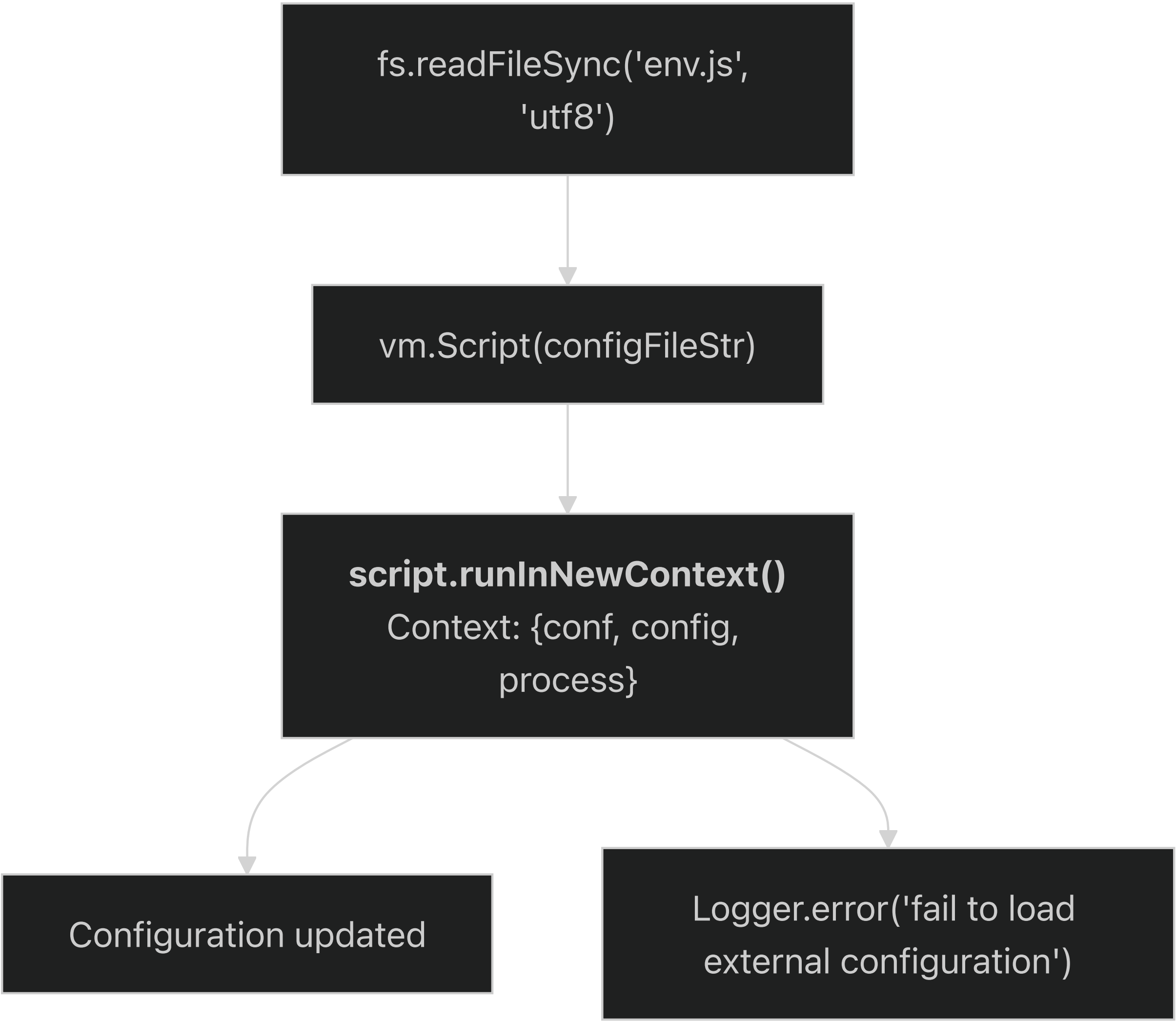
`src/config.ts7-39 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L7-L39) defines these primary sections:

Section	Purpose	Key Settings
Cluster Identity	Application identification	`clusterId`, `clientId`
Kafka Integration	Message queue configuration	`kafkaUrls`, timeout options
Logging	Log output configuration	Console and file appenders
Lotte WebSocket	External service endpoint	`websocketAddress`
Timing Control	Operational intervals	Ping cycles, check intervals

Dynamic Configuration Loading

The system supports runtime configuration overrides through VM script execution in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L41-L53)

`src/config.ts41-53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L41-L53):



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L7-L70)

`src/config.ts7-70 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L7-L70)

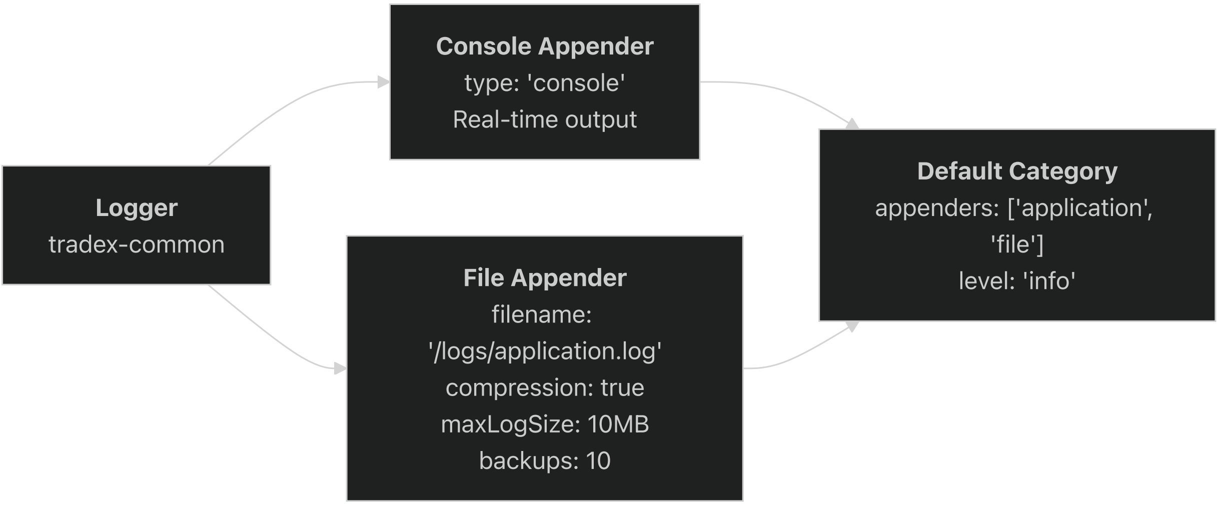
Logging Infrastructure

The logging system utilizes the `tradex-common` Logger component with configurable appenders supporting both console output and file-based persistence.

Logging Configuration

The logging system is configured in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L18-L28)

`src/config.ts18-28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L18-L28) with a dual-appender approach:



Logging Initialization Sequence

The logging system is initialized early in the application lifecycle in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L5-L6)

`src/index.ts5-6 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L5-L6):

1. `Logger.create(config.logger.config, true)` - Initialize with configuration
2. `Logger.info('Starting...')` - First log message indicating startup

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L18-L28)

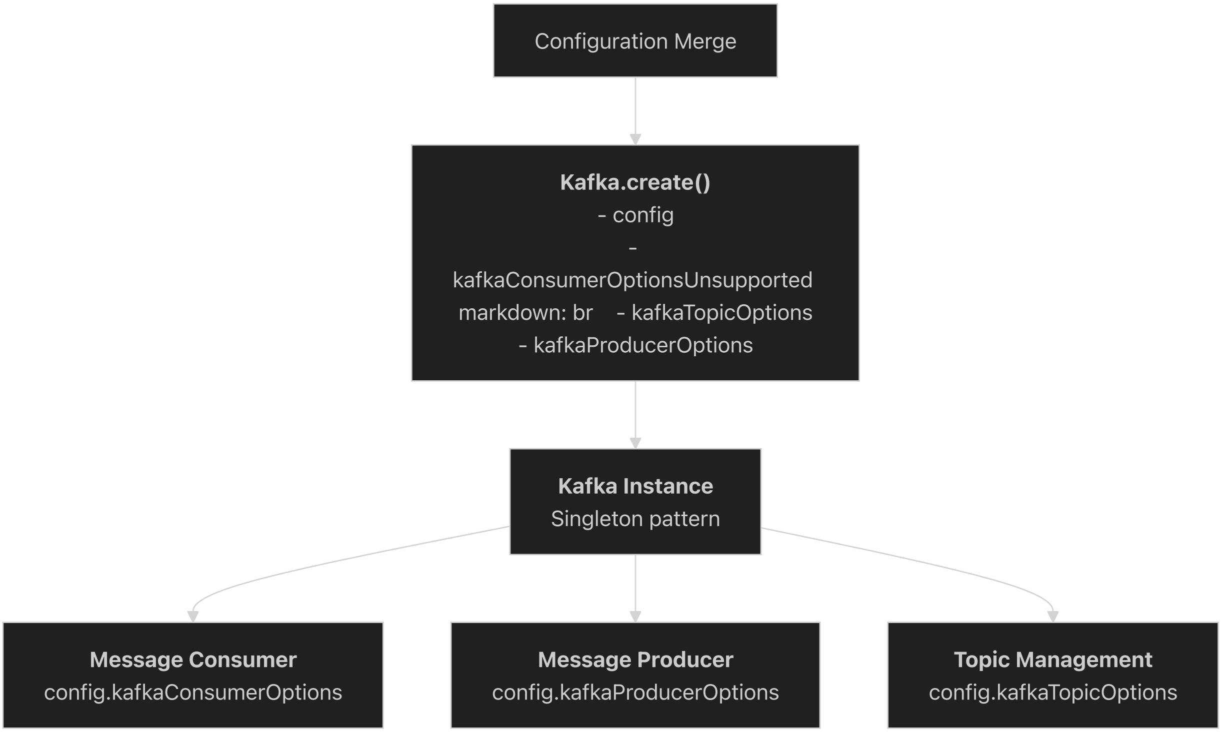
`src/config.ts18-28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L18-L28)[

`src/index.ts5-6](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/`src/index.ts#L5-L6 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L5-L6))

Kafka Integration

The Kafka integration provides the messaging backbone for the application, handling both message consumption and production through the `tradex-common` Kafka component.

Kafka Component Architecture



Configuration Merging Strategy

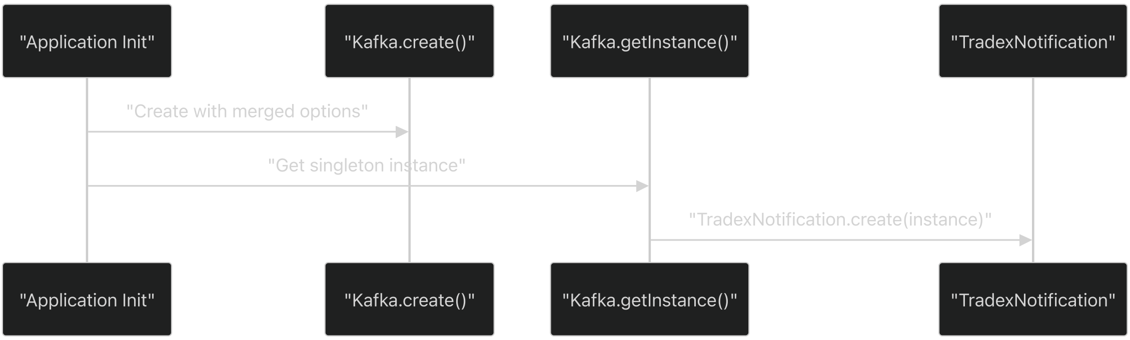
The Kafka configuration follows a hierarchical merging pattern in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L60-L67)

`src/config.ts60-67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L60-L67):

The merging logic applies common options as base configuration, then overlays specific consumer/producer options.

Integration with Notification System

The Kafka instance directly supports the notification system:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L9-L10)

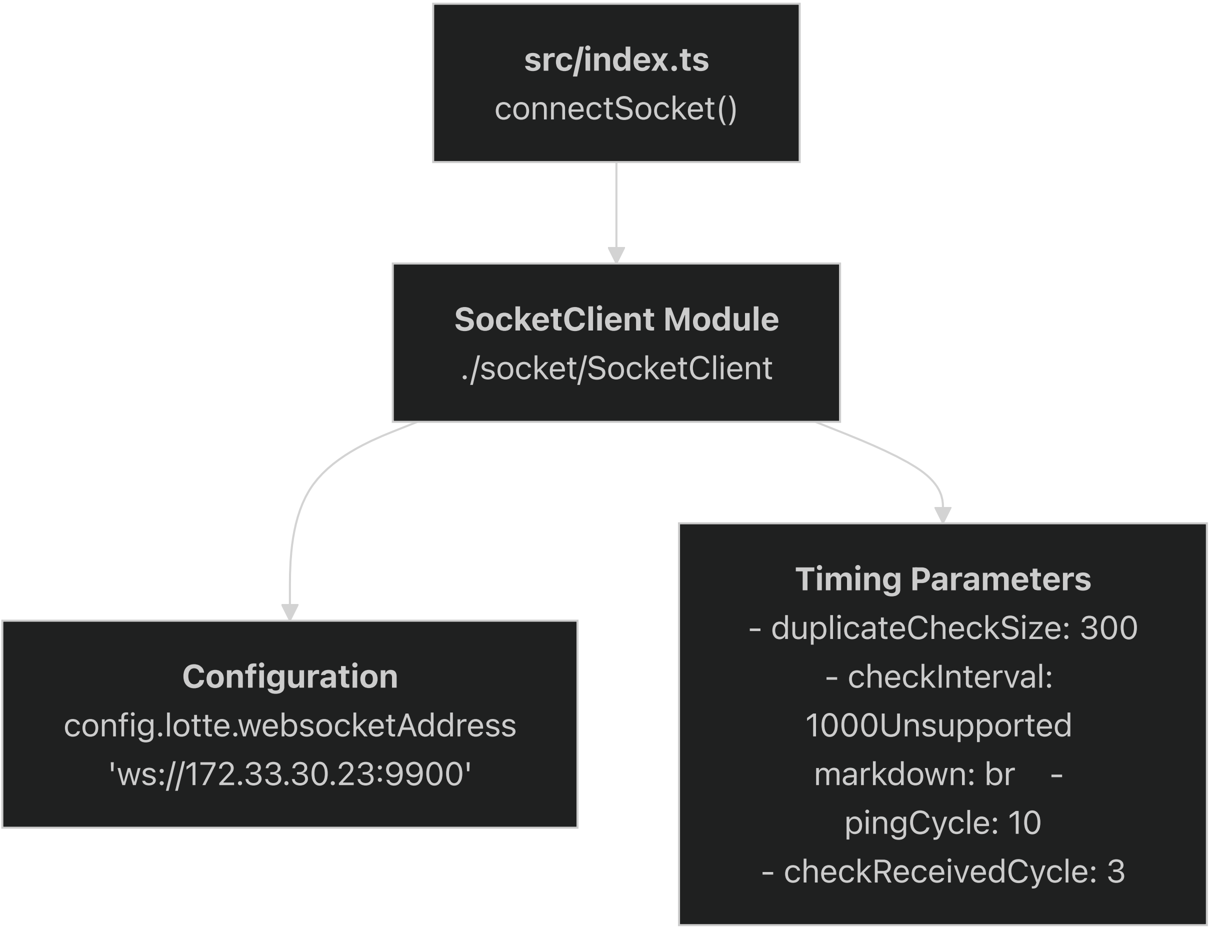
`src/index.ts9-10 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L9-L10)[

`src/config.ts60-67](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/`src/config.ts#L60-L67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L60-L67))

Socket Client Integration

The socket client component handles WebSocket connectivity to external Lotte services, providing the primary bridge functionality of the application.

Socket Client Architecture



Socket Client Configuration

The socket client utilizes specific configuration parameters from (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L29-L38)

`src/config.ts29-38 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L29-L38):

Parameter	Value	Purpose
`websocketAddress`	`ws://172.33.30.23:9900`	Target WebSocket endpoint
`duplicateCheckSize`	300	Message deduplication buffer size
`checkInterval`	1000ms	Health check frequency
`pingCycle`	10	Ping message interval
`checkReceivedCycle`	3	Received message validation cycle

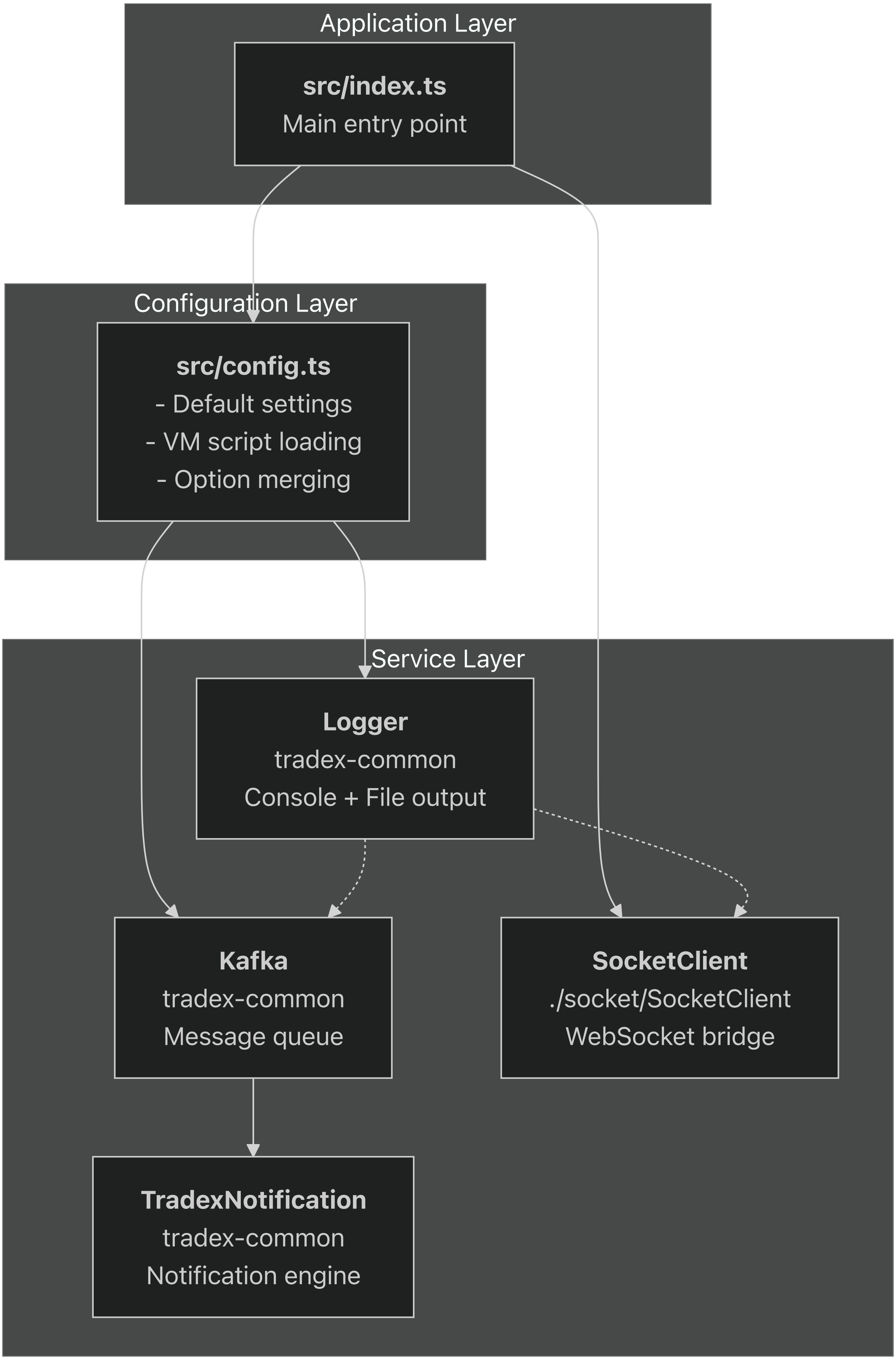
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L11-L11) `src/index.ts11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L11-L11)[

`src/config.ts29-38](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/`src/config.ts#L29-L38 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L29-L38))

Component Integration Flow

The core components work together in a coordinated initialization and runtime flow that establishes the complete bridge functionality.

System Component Interaction



Runtime Dependencies

The component initialization follows strict ordering requirements:

1. **Configuration** must be available before all other components
2. **Logger** must be initialized before any logging occurs
3. **Kafka** must be created before TradexNotification
4. **Socket Client** starts last, after all infrastructure is ready

Shared Configuration Usage

Multiple components consume different sections of the unified configuration object:

- `Logger` uses `config.logger.config`
- `Kafka` uses merged kafka options and URLs
- `SocketClient` uses `config.lotte` and timing parameters
- All components share cluster identification settings

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L1-L20)

`src/index.ts1-20 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/index.ts#L1-L20)[

`src/config.ts1-70](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/`src/config.ts#L1-L70 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/%60src/config.ts#L1-L70))

Data Models

Overview

Relevant source files

- [

  src/model/notification/AdvancePaymentSecuritiesSales.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts))

- [

  src/model/notification/AnnouncementBuyOrderMatching.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts))

- [

  src/model/notification/AnnouncementSellOrderMatching.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts))

Purpose and Scope

This document provides comprehensive documentation of the data models used throughout the lotte-ws-bridge system. The data models are organized into two primary categories: notification models that define the structure for various financial and system events, and request models that define the interface for incoming API calls and data validation.

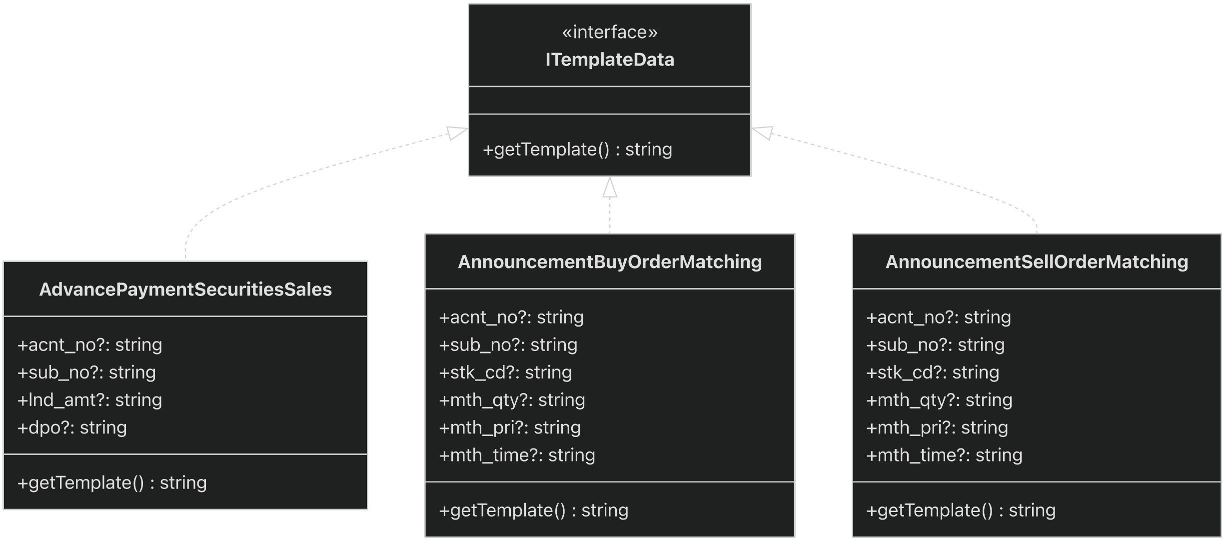
The notification models implement a standardized template-based architecture for generating user-facing notifications across different financial services. For detailed implementation of specific notification types, see Notification Models. For API request validation structures, see Request Models.

Data Model Architecture

The lotte-ws-bridge system employs a sophisticated data modeling approach centered around the `TradexNotification.ITemplateData` interface from the `tradex-common` package. This interface establishes a contract for all notification-related data structures, ensuring consistent template resolution and data handling across the system.

Core Interface Pattern

All notification models implement the `TradexNotification.ITemplateData` interface, which requires a `getTemplate()` method that returns the corresponding template identifier. This pattern enables the notification system to dynamically resolve templates based on the data model type.



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)

src/model/notification/AdvancePaymentSecuritiesSales.ts1-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)[

src/model/notification/AnnouncementBuyOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14))

Template Naming Convention

The system employs a consistent template naming convention where all notification templates follow the pattern `lottehpt_*`. This standardization facilitates template resolution and maintains consistency across the notification rendering pipeline.

Model Class	Template Name
`AdvancePaymentSecuritiesSales`	`lottehpt_advance_payment_securities_sales`
`AnnouncementBuyOrderMatching`	`lottehpt_announcement_buy_order_matching`
`AnnouncementSellOrderMatching`	`lottehpt_announcement_sell_order_matching`

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L9-L11)

src/model/notification/AdvancePaymentSecuritiesSales.ts9-11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L9-L11)[

src/model/notification/AnnouncementBuyOrderMatching.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L11-L13)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L11-L13)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L11-L13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L11-L13))

Data Model Structure Patterns

Field Design Patterns

The notification models follow consistent patterns in their field definitions:

- **Optional Fields**: All data fields are optional (`?`) to accommodate varying data availability across different notification scenarios
- **String Types**: All fields use string types, providing flexibility for different data formats and avoiding type conversion issues
- **Common Fields**: Many models share common field patterns like `acnt_no` (account number) and `sub_no` (sub-account number)

Account Identification Pattern

A recurring pattern across notification models is the account identification structure:

```
acnt_no?: string;  // Primary account number
sub_no?: string;   // Sub-account number
```

This pattern appears in financial transaction notifications and provides consistent account reference across different notification types.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L4-L5)

src/model/notification/AdvancePaymentSecuritiesSales.ts4-5 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L4-L5)[

src/model/notification/AnnouncementBuyOrderMatching.ts4-5](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L4-L5)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L4-L5)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts4-5](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L4-L5 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L4-L5))

Trading-Specific Pattern

Trading-related notifications implement a specialized pattern for order matching information:

```
stk_cd?: string;     // Stock code
mth_qty?: string;    // Matched quantity
mth_pri?: string;    // Matched price
mth_time?: string;   // Match time
```

This pattern standardizes trading event data across buy and sell order notifications.

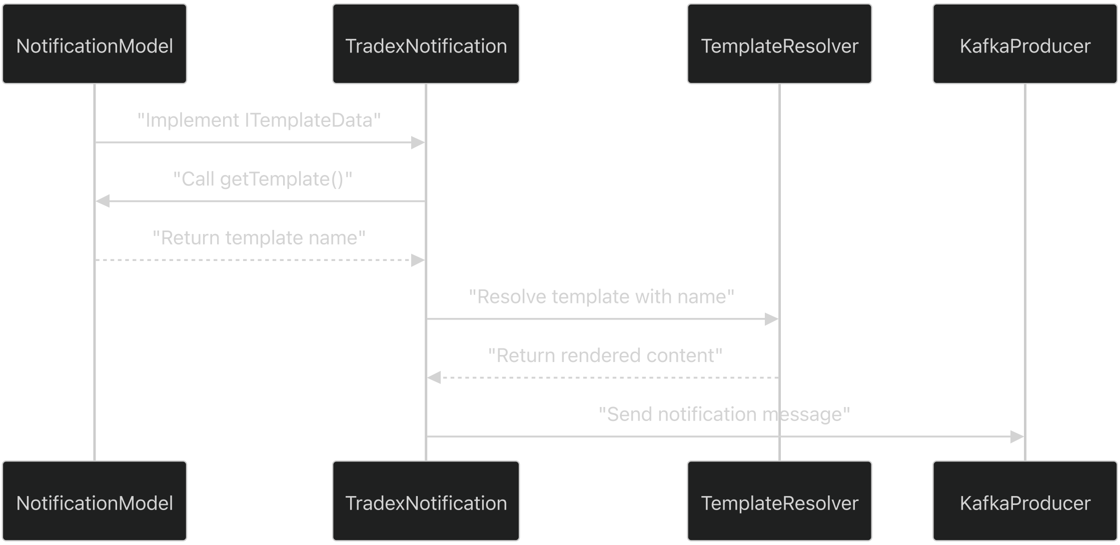
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L6-L9)

src/model/notification/AnnouncementBuyOrderMatching.ts6-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L6-L9)[

src/model/notification/AnnouncementSellOrderMatching.ts6-9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L6-L9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L6-L9))

Integration with Notification System

The data models integrate seamlessly with the broader notification architecture through the template resolution mechanism. The workflow follows this pattern:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)

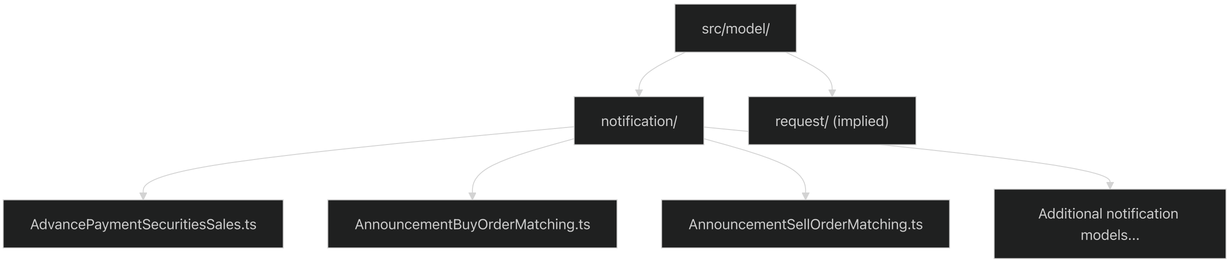
src/model/notification/AdvancePaymentSecuritiesSales.ts1-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)[

src/model/notification/AnnouncementBuyOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14))

Model Organization Structure

The data models are organized in a hierarchical directory structure under `src/model/` that separates concerns and facilitates maintenance:



This organization enables clear separation between different model types while maintaining logical grouping of related functionality.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)

src/model/notification/AdvancePaymentSecuritiesSales.ts1-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)[

src/model/notification/AnnouncementBuyOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14))

Notification Models

Relevant source files

- [

  src/model/notification/AdvancePaymentSecuritiesSales.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts))

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  src/model/notification/AnnouncementBuyOrderMatching.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts))

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  src/model/notification/AnnouncementSellOrderMatching.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts))

- [

  src/model/notification/CancelBuyIssueStock.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts))

- [

  src/model/notification/ChangePassword.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts))

- [

  src/model/notification/DepositStock.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts))

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  src/model/notification/DividendPayment.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts))

- [

  src/model/notification/MarginDisbursement.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts))

Purpose and Scope

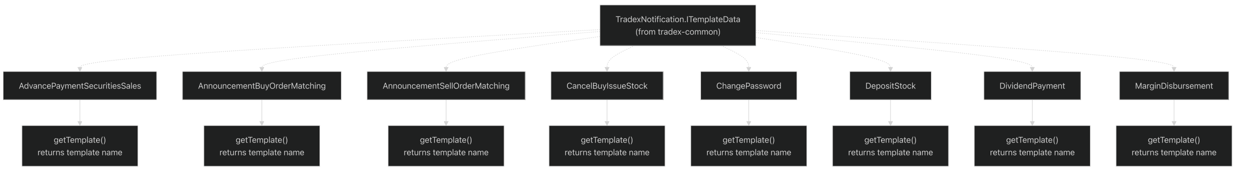
This document covers the notification data models used throughout the lotte-ws-bridge system to represent various financial events and user communications. These models define the data structures for notifications sent to users regarding trading activities, financial transactions, and account operations.

The notification models implement a standardized interface from the `tradex-common` library and integrate with the template-based notification engine. For information about the notification processing engine and Kafka integration, see Core Application Components. For details about specific notification categories, see Trading Notifications, Financial Transaction Notifications, and Account & System Notifications.

Common Notification Interface

All notification models implement the `TradexNotification.ITemplateData` interface from the `tradex-common` library. This interface standardizes how notification data is structured and processed throughout the system.

Interface Implementation Pattern



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)

src/model/notification/AdvancePaymentSecuritiesSales.ts1-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L1-L13)[

src/model/notification/AnnouncementBuyOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L1-L14)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L1-L14)%5B)

src/model/notification/CancelBuyIssueStock.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts#L1-L14)%5B) src/model/notification/ChangePassword.ts1-9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L1-L9)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L1-L9)%5B)

src/model/notification/DepositStock.ts1-12](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L1-L12)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L1-L12)%5B)

src/model/notification/DividendPayment.ts1-14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L1-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L1-L14)%5B)

src/model/notification/MarginDisbursement.ts1-12](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L1-L12 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L1-L12))

Template Naming Convention

All notification models follow a consistent template naming pattern where the `getTemplate()` method returns template identifiers prefixed with `lottehpt_`:

Model Class	Template Name
`AdvancePaymentSecuritiesSales`	`lottehpt_advance_payment_securities_sales`
`AnnouncementBuyOrderMatching`	`lottehpt_announcement_buy_order_matching`
`AnnouncementSellOrderMatching`	`lottehpt_announcement_sell_order_matching`
`CancelBuyIssueStock`	`lottehpt_cancel_buy_issue_stock`
`ChangePassword`	`lottehpt_change_password`
`DepositStock`	`lottehpt_deposit_stock`
`DividendPayment`	`lottehpt_dividend_payment`
`MarginDisbursement`	`lottehpt_margin_disbursement`

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L9-L11) src/model/notification/AdvancePaymentSecuritiesSales.ts9-11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L9-L11)[

src/model/notification/AnnouncementBuyOrderMatching.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L11-L13)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L11-L13)%5B)

src/model/notification/AnnouncementSellOrderMatching.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L11-L13)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementSellOrderMatching.ts#L11-L13)%5B)

src/model/notification/CancelBuyIssueStock.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts#L11-L13)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/CancelBuyIssueStock.ts#L11-L13)%5B)

src/model/notification/ChangePassword.ts6-8](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L6-L8)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L6-L8)%5B)

src/model/notification/DepositStock.ts9-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L9-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L9-L11)%5B)

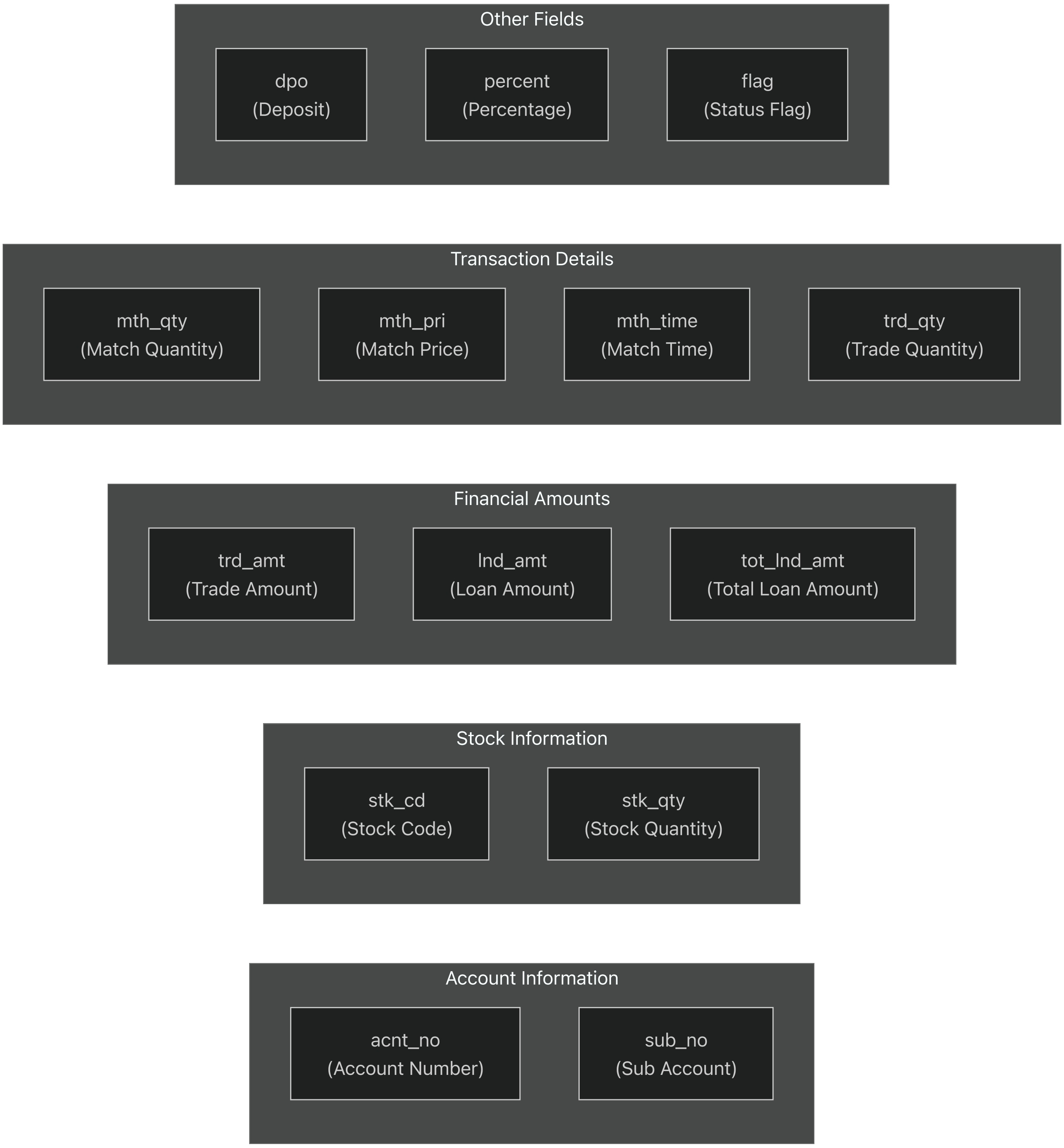
src/model/notification/DividendPayment.ts11-13](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L11-L13)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L11-L13)%5B)

src/model/notification/MarginDisbursement.ts9-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L9-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L9-L11))

Data Structure Patterns

Common Field Types

The notification models utilize consistent field naming and typing patterns across different notification types:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L3-L7)

src/model/notification/AdvancePaymentSecuritiesSales.ts3-7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L3-L7)[

src/model/notification/AnnouncementBuyOrderMatching.ts3-9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L3-L9)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L3-L9)%5B) src/model/notification/DividendPayment.ts3-9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L3-L9)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DividendPayment.ts#L3-L9)%5B)

src/model/notification/MarginDisbursement.ts3-7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L3-L7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/MarginDisbursement.ts#L3-L7))

Field Usage by Category

Field Name	Trading	Financial	Account/System	Description
`acnt_no`	✓	✓	-	Account number identifier
`sub_no`	✓	✓	-	Sub-account identifier
`stk_cd`	✓	✓	-	Stock symbol/code
`stk_qty`	✓	-	-	Stock quantity
`trd_amt`	✓	✓	-	Transaction amount
`mth_qty`	✓	-	-	Matched quantity
`mth_pri`	✓	-	-	Matched price
`mth_time`	✓	-	-	Match timestamp
`lnd_amt`	-	✓	-	Loan amount
`dpo`	✓	✓	-	Deposit information
`percent`	-	✓	-	Percentage value
`flag`	-	-	✓	Status flag

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L4-L9)

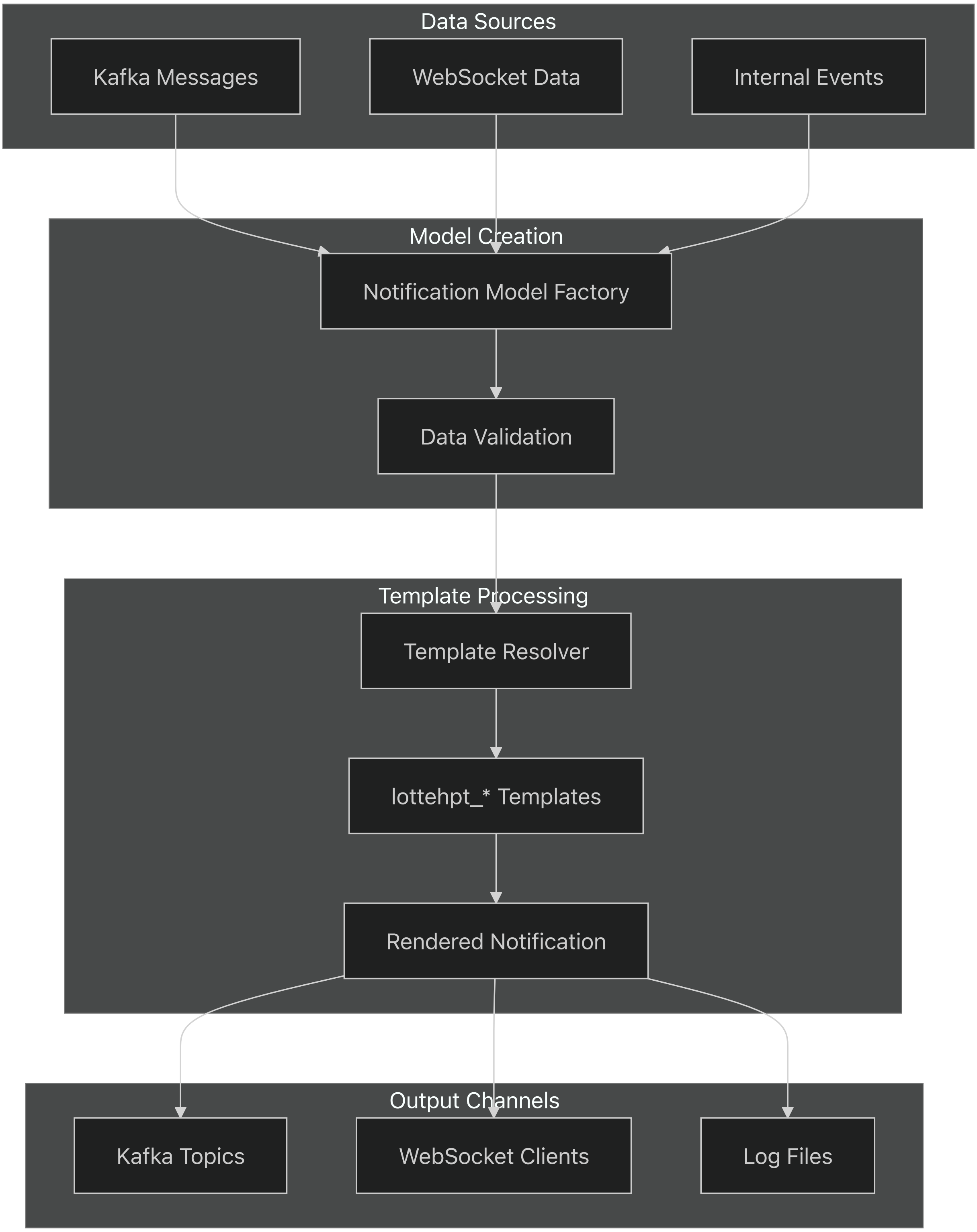
src/model/notification/AnnouncementBuyOrderMatching.ts4-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AnnouncementBuyOrderMatching.ts#L4-L9)[

src/model/notification/AdvancePaymentSecuritiesSales.ts4-7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L4-L7)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/AdvancePaymentSecuritiesSales.ts#L4-L7)%5B)

src/model/notification/ChangePassword.ts4](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L4-L4)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/ChangePassword.ts#L4-L4)%5B)

src/model/notification/DepositStock.ts4-7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L4-L7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/DepositStock.ts#L4-L7))

Notification Data Flow



Sources: All notification model files (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/*.ts)

src/model/notification/*.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/notification/*.ts)

The notification models serve as the data bridge between raw event data and template-rendered user notifications, ensuring consistent structure and reliable template resolution throughout the lotte-ws-bridge system.

Trading Notifications

Relevant source files

- [

  src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts))

- [

  src/model/request/IAnnouncementBuyOrderMatchingRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts))

- [

  src/model/request/IAnnouncementSellOrderMatchingRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts))

- [

  src/model/request/ICancelBuyIssueStockRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts))

- [

  src/model/request/IChangePasswordRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts))

This document covers the request interface definitions used by the lotte-ws-bridge system for processing incoming API calls and data validation. These TypeScript interfaces define the structure of data received from external systems and WebSocket clients.

For information about the corresponding notification data models that are generated from these requests, see Notification Models. For details about the core application components that process these requests, see Core Application Components.

Purpose and Scope

The request models in this system define standardized data structures for various financial operations and account management activities. Each interface specifies required fields and data types for incoming requests, ensuring consistent data validation and processing throughout the bridge application.

All request models follow a common pattern with shared base fields while extending to include operation-specific data. These models are located in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/)

src/model/request/ (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/) and serve as the input contracts for the notification generation system.

Common Request Structure

All request interfaces share a set of core fields that provide event identification and account context:

Field	Type	Purpose
`date`	string	Event date timestamp
`event_seqno`	string	Unique sequence number for the event
`event_code`	string	Event type classification code
`acnt_no`	string	Account number

Additional fields vary based on the specific request type and business logic requirements.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts1-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)[

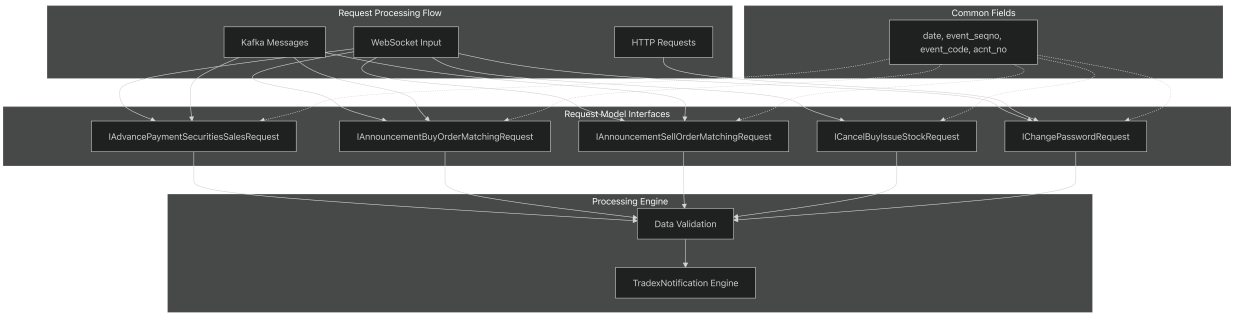
src/model/request/IAnnouncementBuyOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)%5B)

src/model/request/IAnnouncementSellOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11)%5B)

src/model/request/ICancelBuyIssueStockRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)%5B)

src/model/request/IChangePasswordRequest.ts1-7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7))

Request Model Architecture



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts1-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)[

src/model/request/IAnnouncementBuyOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)%5B)

src/model/request/IAnnouncementSellOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-

bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11)[
(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11)%5B)

src/model/request/ICancelBuyIssueStockRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)[
(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)%5B)

src/model/request/IChangePasswordRequest.ts1-7](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7))

Trading Operations Requests

Order Matching Requests

The system handles both buy and sell order matching events through specialized request interfaces:

IAnnouncementBuyOrderMatchingRequest

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`stk_cd`	string	Stock code
`mth_qty`	string	Matched quantity
`mth_pri`	string	Matched price
`mth_time`	string	Matching timestamp

IAnnouncementSellOrderMatchingRequest

Identical structure to buy order matching, handling sell-side transactions with the same field definitions.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)

[src/model/request/IAnnouncementBuyOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11) (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)[

src/model/request/IAnnouncementSellOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementSellOrderMatchingRequest.ts#L1-L11))

Order Cancellation Requests

ICancelBuyIssueStockRequest

Handles buy order cancellations for issued stocks:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`trd_amt`	string	Trade amount
`trd_qty`	string	Trade quantity
`stk_cd`	string	Stock code
`dpo`	string	Depository participant organization

Sources: [(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)

[src/model/request/ICancelBuyIssueStockRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11) (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)

Financial Transaction Requests

Securities Sales Advance Payment

IAdvancePaymentSecuritiesSalesRequest

Processes advance payment requests for securities sales:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`lnd_amt`	string	Loan amount
`dpo`	string	Depository participant organization

Sources: [(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

[src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts1-9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9) (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

Account Management Requests

Password Change Requests

IChangePasswordRequest

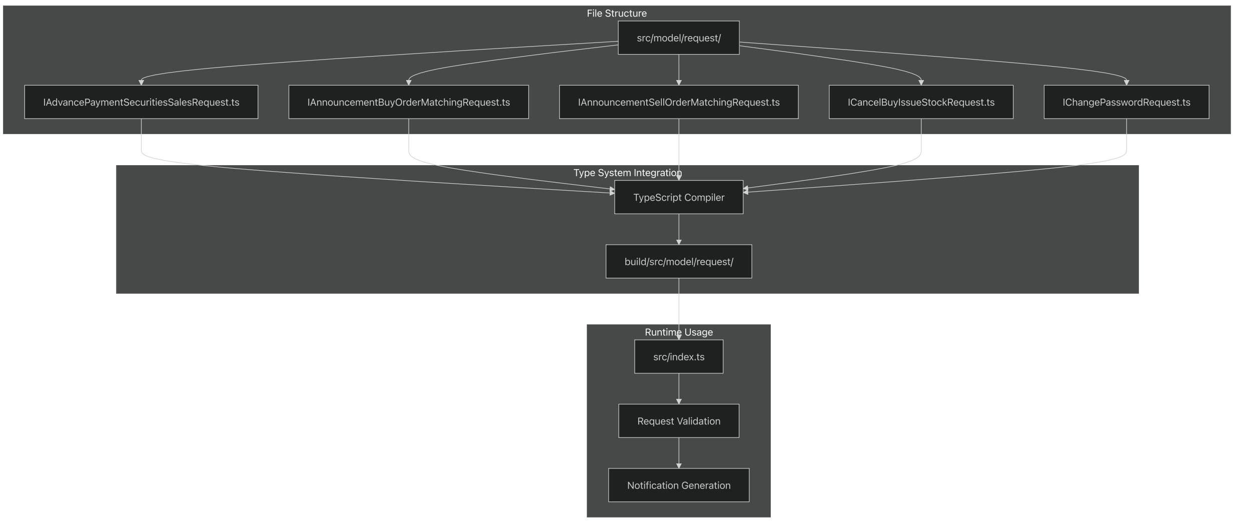
Handles password modification events:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`flag`	string	Operation flag

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)

src/model/request/IChangePasswordRequest.ts1-7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)

Request Model Integration



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts1-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)[

src/model/request/IAnnouncementBuyOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)[

(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)%5B)

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src/model/request/ICancelBuyIssueStockRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)%5B)

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Financial Transaction Notifications

Relevant source files

- [

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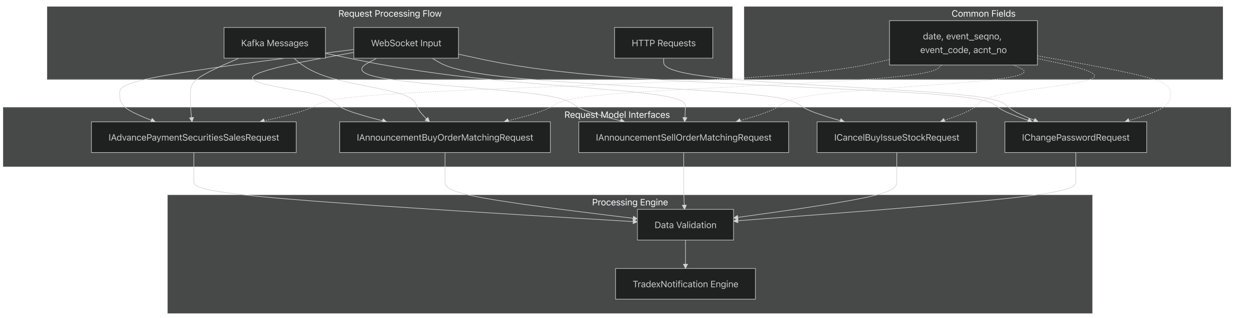
Request Model Architecture



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Trading Operations Requests

Order Matching Requests

The system handles both buy and sell order matching events through specialized request interfaces:

IAnnouncementBuyOrderMatchingRequest

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`stk_cd`	string	Stock code
`mth_qty`	string	Matched quantity
`mth_pri`	string	Matched price
`mth_time`	string	Matching timestamp

IAnnouncementSellOrderMatchingRequest

Identical structure to buy order matching, handling sell-side transactions with the same field definitions.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)

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Order Cancellation Requests

ICancelBuyIssueStockRequest

Handles buy order cancellations for issued stocks:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`trd_amt`	string	Trade amount
`trd_qty`	string	Trade quantity
`stk_cd`	string	Stock code
`dpo`	string	Depository participant organization

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)

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Financial Transaction Requests

Securities Sales Advance Payment

IAdvancePaymentSecuritiesSalesRequest

Processes advance payment requests for securities sales:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`lnd_amt`	string	Loan amount
`dpo`	string	Depository participant organization

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

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Account Management Requests

Password Change Requests

IChangePasswordRequest

Handles password modification events:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`flag`	string	Operation flag

Sources: [(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)

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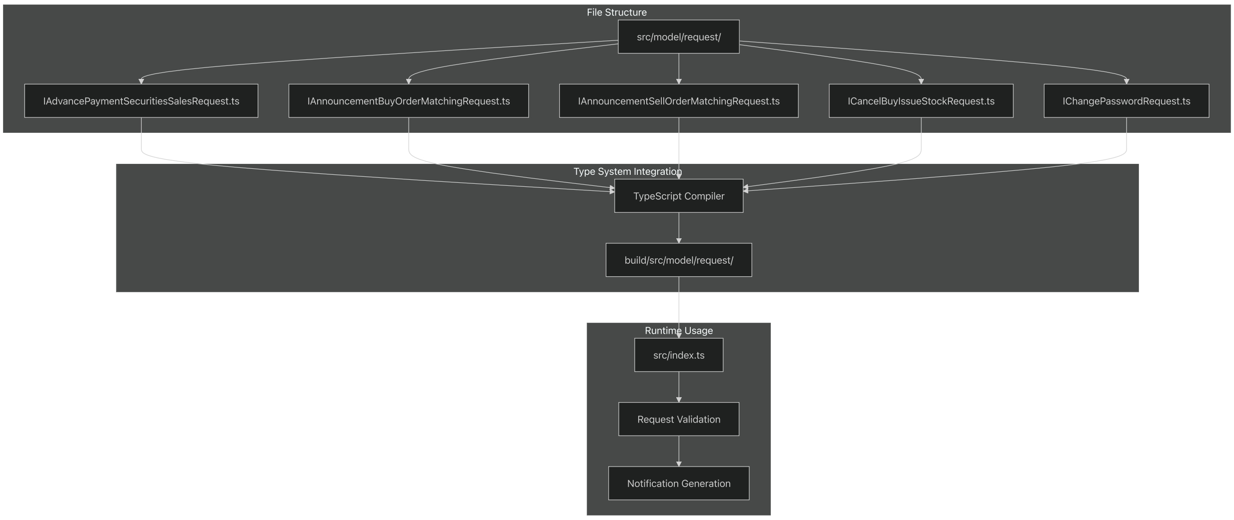

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Account & System Notifications

Relevant source files

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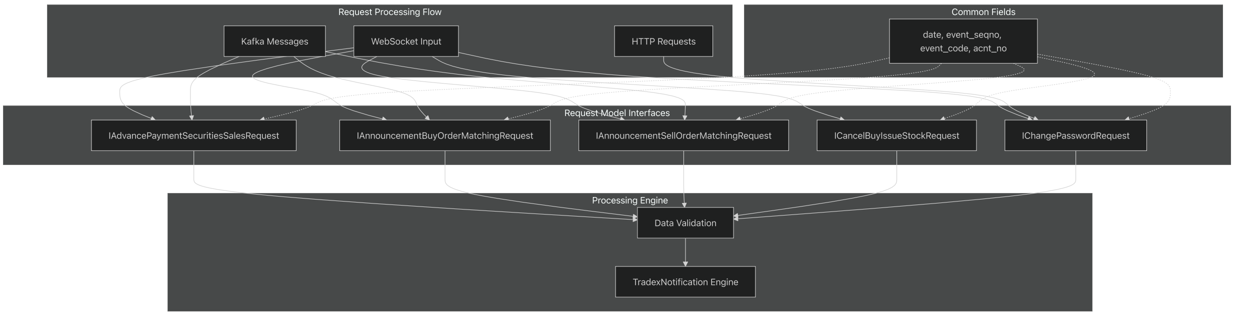

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Trading Operations Requests

Order Matching Requests

The system handles both buy and sell order matching events through specialized request interfaces:

IAnnouncementBuyOrderMatchingRequest

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`stk_cd`	string	Stock code
`mth_qty`	string	Matched quantity
`mth_pri`	string	Matched price
`mth_time`	string	Matching timestamp

IAnnouncementSellOrderMatchingRequest

Identical structure to buy order matching, handling sell-side transactions with the same field definitions.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11) [src/model/request/IAnnouncementBuyOrderMatchingRequest.ts1-11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)[

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Order Cancellation Requests

ICancelBuyIssueStockRequest

Handles buy order cancellations for issued stocks:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`trd_amt`	string	Trade amount
`trd_qty`	string	Trade quantity
`stk_cd`	string	Stock code
`dpo`	string	Depository participant organization

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/ICancelBuyIssueStockRequest.ts#L1-L11)

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Financial Transaction Requests

Securities Sales Advance Payment

IAdvancePaymentSecuritiesSalesRequest

Processes advance payment requests for securities sales:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`lnd_amt`	string	Loan amount
`dpo`	string	Depository participant organization

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

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Account Management Requests

Password Change Requests

IChangePasswordRequest

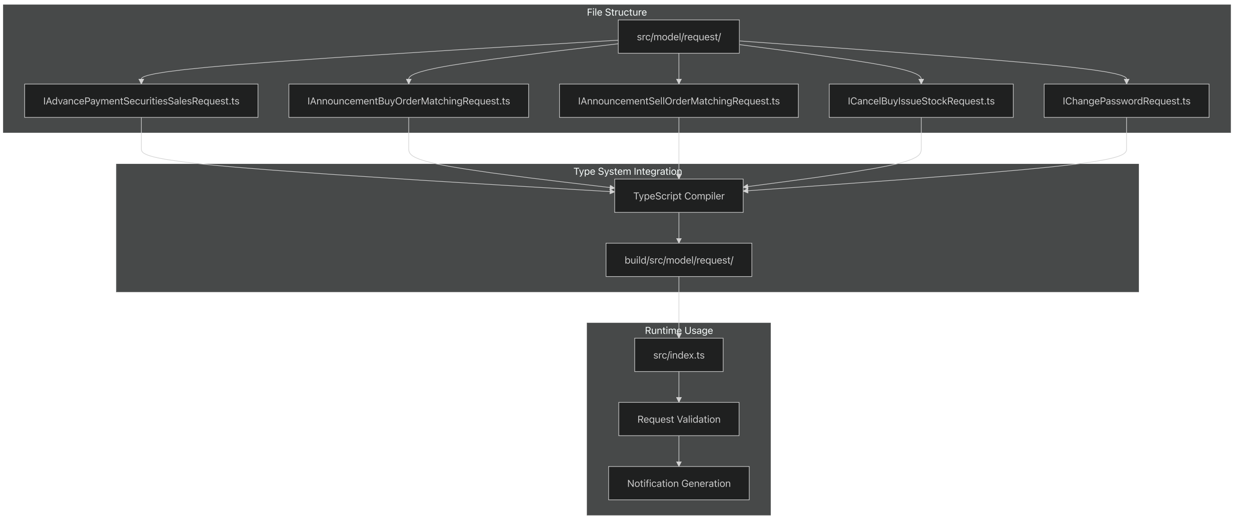
Handles password modification events:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`flag`	string	Operation flag

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)

src/model/request/IChangePasswordRequest.ts1-7 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IChangePasswordRequest.ts#L1-L7)

Request Model Integration



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Request Models

Relevant source files

- [

  src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts))
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This document covers the request interface definitions used by the lotte-ws-bridge system for processing incoming API calls and data validation. These TypeScript interfaces define the structure of data received from external systems and WebSocket clients.

For information about the corresponding notification data models that are generated from these requests, see Notification Models. For details about the core application components that process these requests, see Core Application Components.

Purpose and Scope

The request models in this system define standardized data structures for various financial operations and account management activities. Each interface specifies required fields and data types for incoming requests, ensuring consistent data validation and processing throughout the bridge application.

All request models follow a common pattern with shared base fields while extending to include operation-specific data. These models are located in (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/)

src/model/request/ (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/) and serve as the input contracts for the notification generation system.

Common Request Structure

All request interfaces share a set of core fields that provide event identification and account context:

Field	Type	Purpose
`date`	string	Event date timestamp
`event_seqno`	string	Unique sequence number for the event
`event_code`	string	Event type classification code
`acnt_no`	string	Account number

Additional fields vary based on the specific request type and business logic requirements.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts1-9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)[

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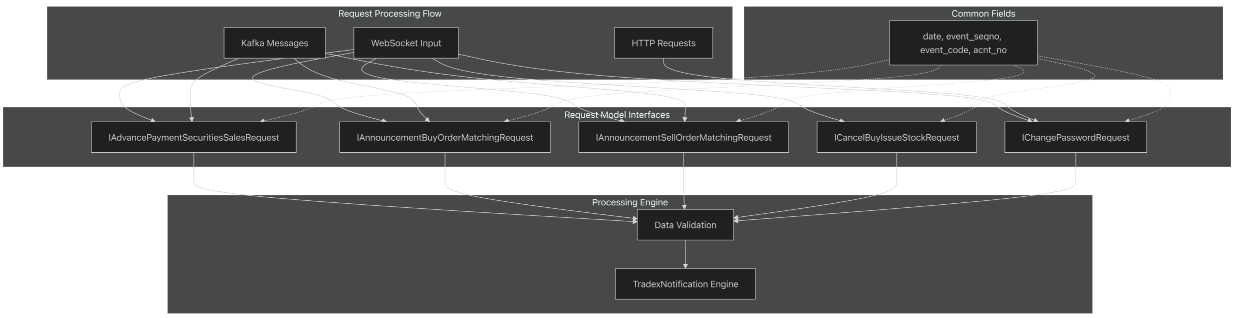
Request Model Architecture



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Trading Operations Requests

Order Matching Requests

The system handles both buy and sell order matching events through specialized request interfaces:

IAnnouncementBuyOrderMatchingRequest

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`stk_cd`	string	Stock code
`mth_qty`	string	Matched quantity
`mth_pri`	string	Matched price
`mth_time`	string	Matching timestamp

IAnnouncementSellOrderMatchingRequest

Identical structure to buy order matching, handling sell-side transactions with the same field definitions.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAnnouncementBuyOrderMatchingRequest.ts#L1-L11)

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Order Cancellation Requests

ICancelBuyIssueStockRequest

Handles buy order cancellations for issued stocks:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`trd_amt`	string	Trade amount
`trd_qty`	string	Trade quantity
`stk_cd`	string	Stock code
`dpo`	string	Depository participant organization

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Financial Transaction Requests

Securities Sales Advance Payment

IAdvancePaymentSecuritiesSalesRequest

Processes advance payment requests for securities sales:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`sub_no`	string	Sub-account number
`lnd_amt`	string	Loan amount
`dpo`	string	Depository participant organization

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/model/request/IAdvancePaymentSecuritiesSalesRequest.ts#L1-L9)

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Account Management Requests

Password Change Requests

IChangePasswordRequest

Handles password modification events:

Field	Type	Description
`date`	string	Event date
`event_seqno`	string	Event sequence number
`event_code`	string	Event classification
`acnt_no`	string	Account number
`flag`	string	Operation flag

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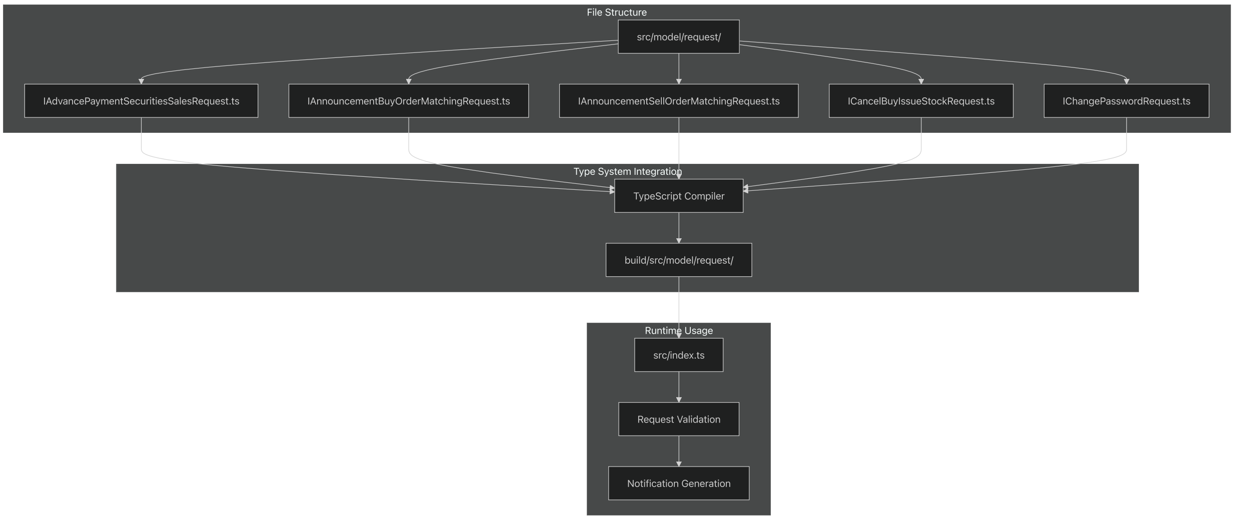

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External Integrations

Relevant source files

- [

  package-lock.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json))

- [

  src/config.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts))

- [

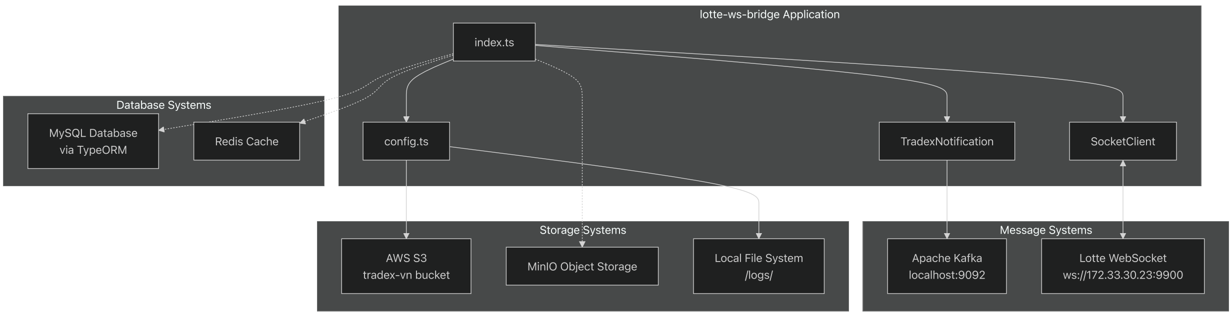
  src/index.ts](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts))

This document describes how the `lotte-ws-bridge` system integrates with external services and systems. It covers the configuration, initialization, and operational aspects of connections to message queues, WebSocket services, storage systems, and databases.

For information about the core application components and internal architecture, see Core Application Components. For data model definitions used in these integrations, see Data Models.

Integration Overview

The `lotte-ws-bridge` acts as a bridge between multiple external systems, handling message routing, notification processing, and data persistence across various service boundaries.



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)

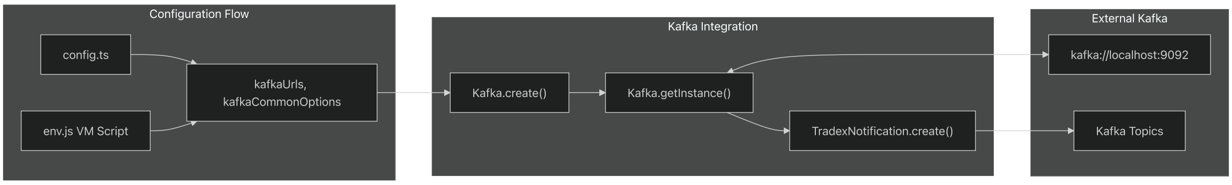
src/index.ts1-19 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L1-L19)[

src/config.ts7-39](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L7-L39)%5B) package-lock.json438-474](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L474 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L474))

Message Queue Integration (Kafka)

The system uses Apache Kafka for asynchronous message processing and notification distribution through the `tradex-common` library.

Kafka Configuration



Configuration Key	Default Value	Purpose
`kafkaUrls`	`['localhost:9092']`	Kafka broker endpoints
`kafkaCommonOptions`	`{}`	Shared Kafka client options
`kafkaConsumerOptions`	`{}`	Consumer-specific options
`kafkaProducerOptions`	`{}`	Producer-specific options
`kafkaTopicOptions`	`{}`	Topic configuration options
`defaultKafkaTimeout`	`10000`	Default timeout in milliseconds

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L17)

src/config.ts10-17 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L10-L17)[

src/index.ts9-10](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L10 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L10))

Kafka Integration Code Flow

The Kafka integration follows this initialization pattern:

1. **Configuration Loading**:[

   src/config.ts60-67](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67))

2. **Kafka Instance Creation**:[

   src/index.ts9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L9 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L9-L9))

3. **Notification System Binding**:[

   src/index.ts10](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L10-L10 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L10-L10))

The `kafkaConsumerOptions` and `kafkaProducerOptions` are merged with `kafkaCommonOptions` to create the final configuration objects.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67)

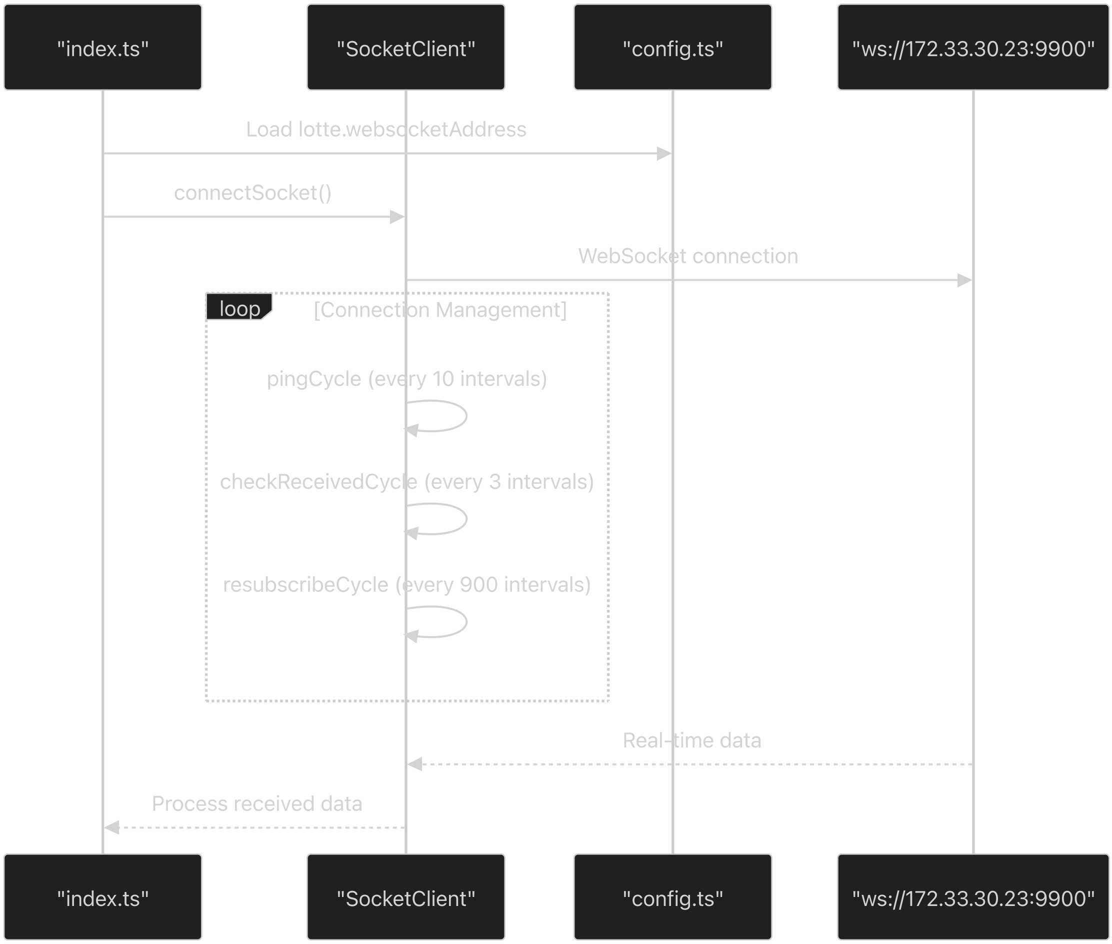
src/config.ts60-67 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L60-L67)[

src/index.ts8-12](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L8-L12))

WebSocket Integration (Lotte)

The system maintains a persistent WebSocket connection to the Lotte service for real-time data exchange.

WebSocket Connection Architecture



WebSocket Configuration Parameters

Parameter	Default Value	Purpose
`websocketAddress`	`'ws://172.33.30.23:9900'`	Lotte WebSocket server URL
`duplicateCheckSize`	`300`	Duplicate message detection buffer
`checkInterval`	`1000`	Health check interval (ms)
`pingCycle`	`10`	Ping frequency cycles
`checkReceivedCycle`	`3`	Received message check cycles
`resubscribeCycle`	`900`	Resubscription cycle (0 to disable)

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L38)

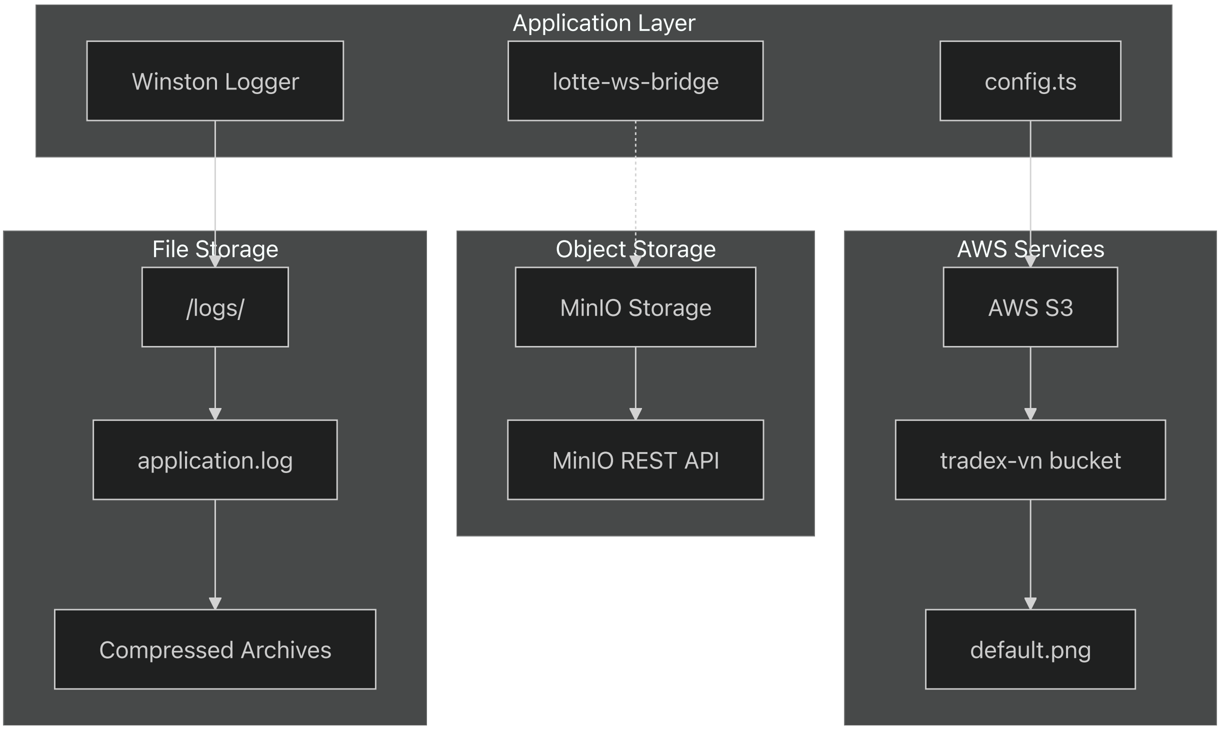
src/config.ts29-38 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L29-L38)[

src/index.ts11](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L11-L11 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/index.ts#L11-L11))

Storage Systems Integration

The application integrates with multiple storage systems for different purposes.

Storage Architecture Overview



AWS S3 Configuration

The system uses AWS S3 for storing default avatar images:

```
defaultAvatar: 'https://s3-ap-southeast-1.amazonaws.com/tradex-vn/avatar/defa
```

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L16-L16)

src/config.ts16 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L16-L16)[

package-lock.json438-474](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L474 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L474))

File System Logging

Log files are stored locally with rotation and compression:

Setting	Value	Purpose
Log Directory	`/logs/`	Log file storage location
Log File	`application.log`	Main application log
Max Log Size	`10485760` bytes	Log rotation trigger
Backup Count	`10`	Number of archived logs
Compression	`true`	Compress archived logs

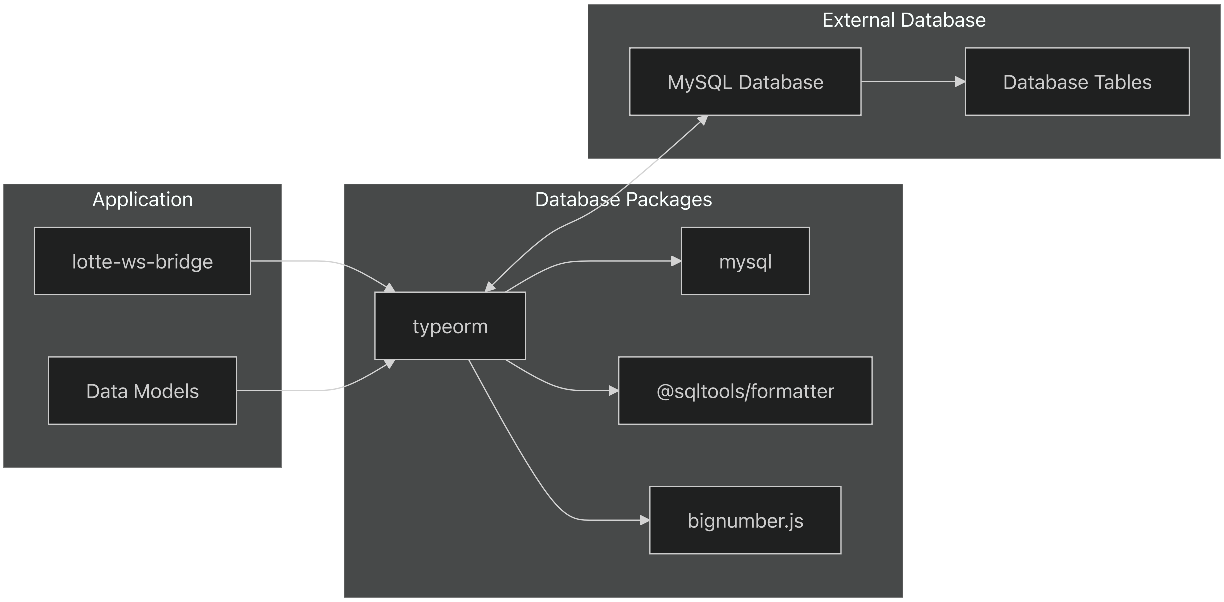
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28)

src/config.ts18-28 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L18-L28)

Database Integration

The system supports database connectivity through TypeORM and related packages.

Database Dependencies



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L91-L95)

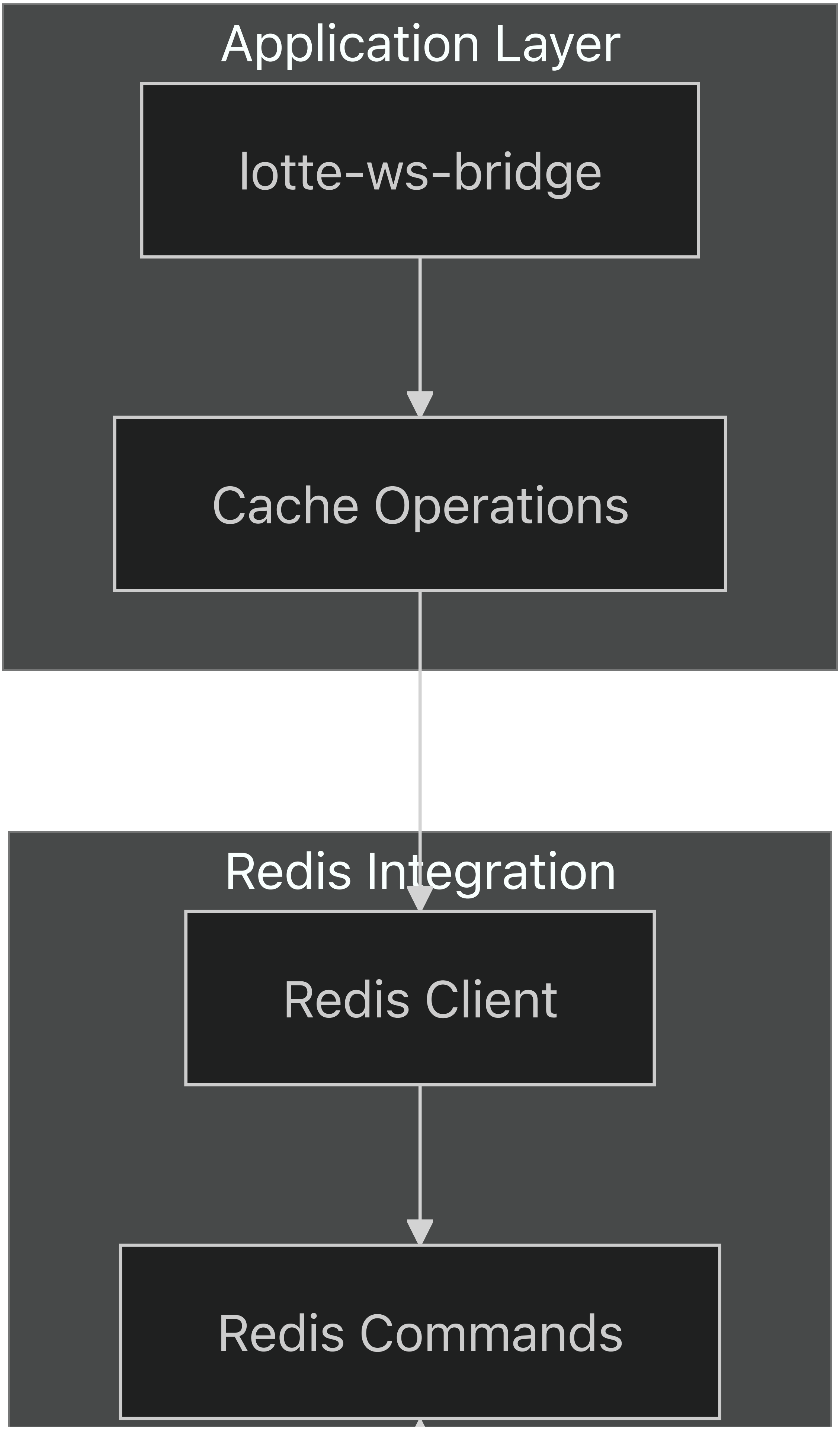
package-lock.json91-95 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L91-L95)[

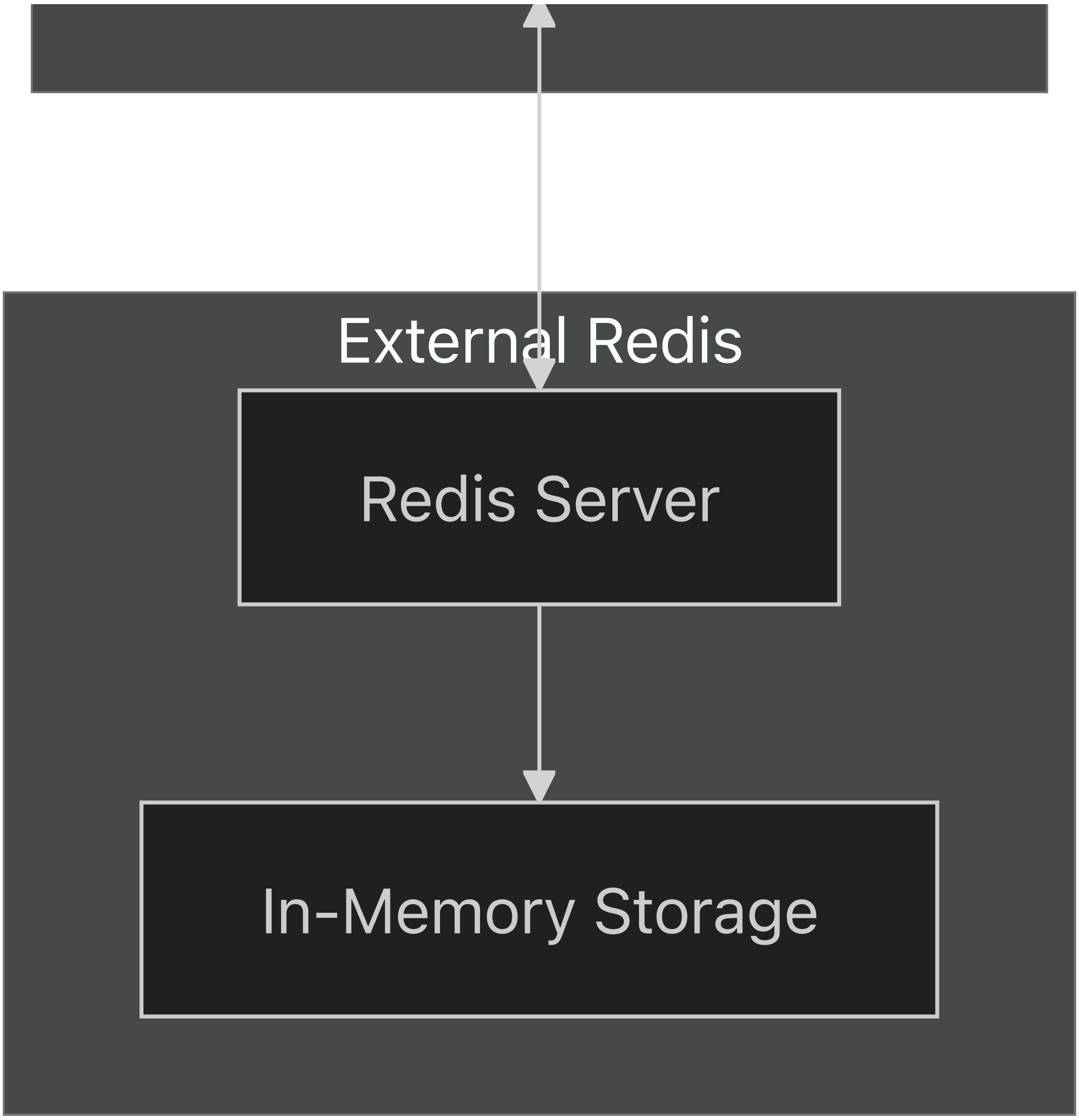
package-lock.json499-503](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L499-L503 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L499-L503))

Cache Integration (Redis)

The system includes Redis client libraries for caching functionality.

Redis Client Architecture


Application Layer
lotte-ws-bridge
Cache Operations
Redis Integration
Redis Client
Redis Commands



The Redis integration is available through the dependency tree but specific configuration is not visible in the provided configuration files.

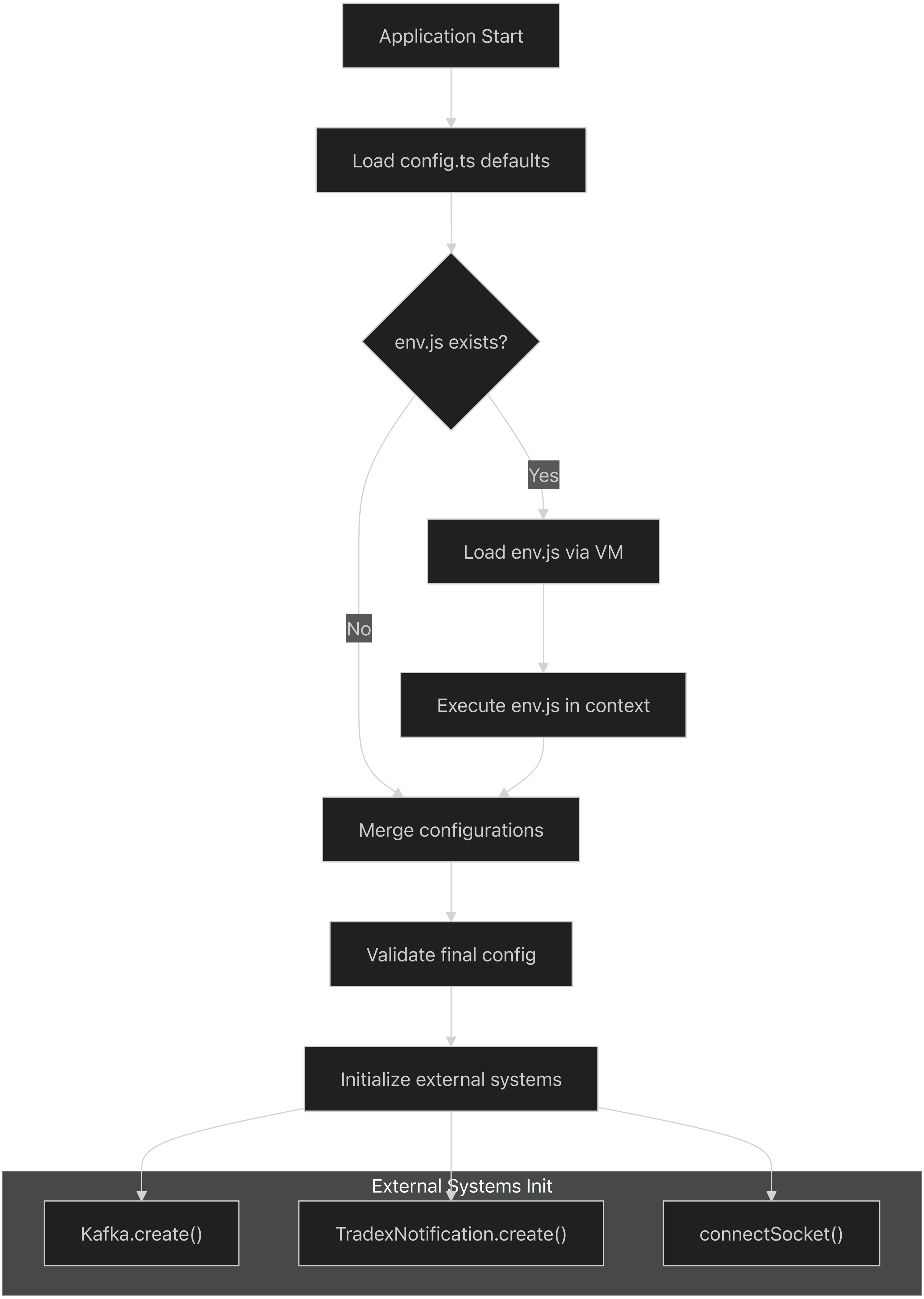
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L1-L5000)

package-lock.json1-5000 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L1-L5000) (Redis dependencies present)

Configuration and Initialization Flow

The application uses a sophisticated configuration system with environment-specific overrides.

Configuration Loading Process



Environment Override Mechanism

The configuration system supports dynamic environment-specific overrides through VM script execution:

```
// env.js example structure
conf.kafkaUrls = ['production-kafka:9092'];
conf.lotte.websocketAddress = 'ws://production-lotte:9900';
```

This approach allows for secure configuration management without exposing sensitive values in the codebase.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)

src/config.ts41-53 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L41-L53)[

src/config.ts55-69](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L69 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/src/config.ts#L55-L69))

Integration Dependencies

The external integrations are supported by these key dependencies:

Dependency	Version Range	Purpose
`aws-sdk`	`^2.1456.0`	AWS services integration
`axios`	`^0.21.4`	HTTP client for REST APIs
`typeorm`	Latest	Database ORM
`winston`	Via tradex-common	Structured logging
`ws`	Via dependencies	WebSocket client

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L482)

package-lock.json438-482 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L438-L482)[

package-lock.json476-482](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L476-L482 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package-lock.json#L476-L482))

Development Workflow

Relevant source files

- [

  .gitignore](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.gitignore (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.gitignore))

- [

  .nvmrc](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc))

- [

  dockerfile](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile))

- [

  package.json](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json))

- [

  prettier.config.js](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js))

Purpose and Scope

This document outlines the development workflow, build processes, code quality standards, and deployment practices for the lotte-ws-bridge system. It covers the complete developer experience from environment setup through production deployment.

For information about system architecture and component relationships, see System Architecture. For configuration management details, see Configuration Management.

Development Environment Setup

Node.js Version Management

The project uses Node.js version `10.16.3` as specified in the `.nvmrc` file. This ensures consistency across development environments and matches the production Docker base image.

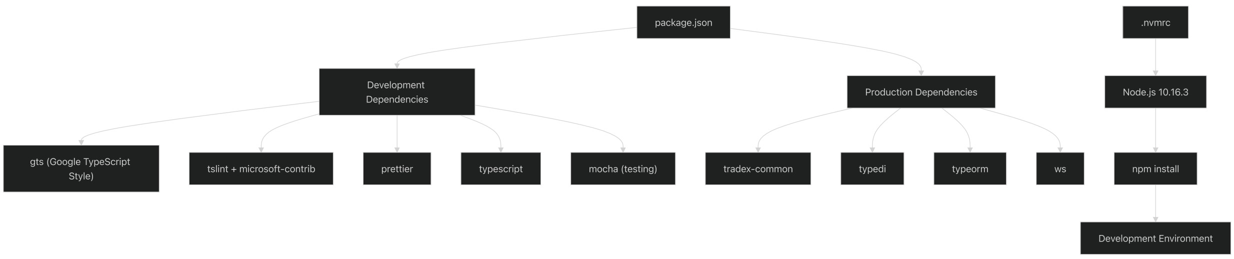
```
nvm use
```

Required Dependencies

The development environment requires several native module dependencies for compilation:

- Python 3
- Make
- GCC compiler (g++)
- Git
- Bash

Development Environment Workflow:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)

.nvmrc1 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.nvmrc#L1-L1)[

package.json22-35](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L22-L35)%5B)

package.json36-55](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L55 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L36-L55))

Code Quality and Standards

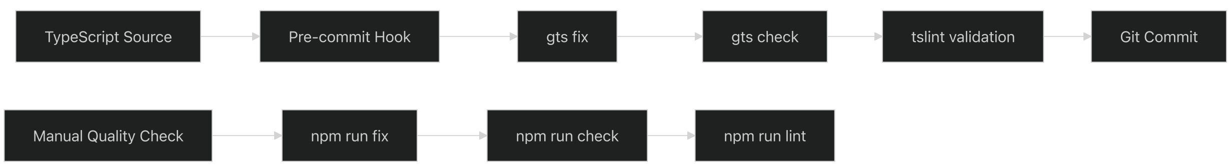
TypeScript Configuration

The project uses TypeScript with strict typing and Google TypeScript Style (GTS) for consistent code formatting and quality.

Linting and Formatting Tools

Tool	Purpose	Configuration
`gts`	Google TypeScript Style guide enforcement	Provides `check` and `fix` commands
`tslint`	TypeScript linting with Microsoft contrib rules	Uses stylish reporter
`prettier`	Code formatting	Custom config with single quotes, 120 char width
`husky`	Git hooks for pre-commit quality checks	Runs `fix` and `check` before commits

Code Quality Workflow:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)

package.json56-60 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)[

package.json9](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L9-L9)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L9-L9)%5B)

package.json14](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L14-L14)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L14-L14)%5B)

package.json16](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L16-L16)[(https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L16-L16)%5B)

prettier.config.js1-8](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js#L1-L8 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js#L1-L8))

Prettier Configuration

The code formatting follows these standards:

- Single quotes for strings
- Trailing commas (ES5 style)
- Arrow function parentheses always required
- Line endings: LF
- Tab width: 2 spaces
- Print width: 120 characters

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js#L1-L8)

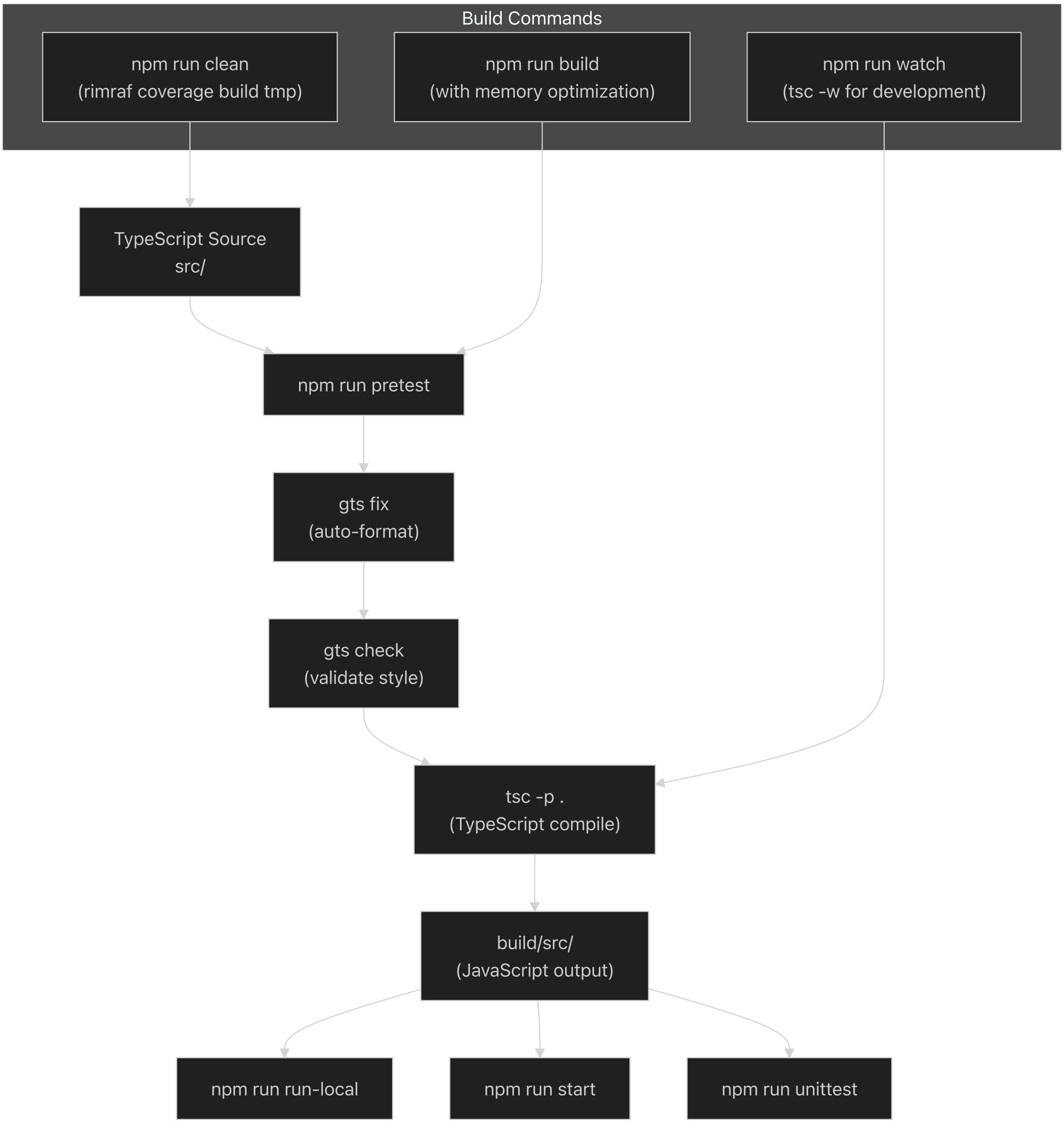
prettier.config.js1-8 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/prettier.config.js#L1-L8)

Build Workflow

TypeScript Compilation Process

The build process follows a multi-stage approach with quality checks integrated:

Build Pipeline:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18)

package.json5-18 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L5-L18)

Available Build Scripts

Script	Purpose	Command
`clean`	Remove build artifacts	`rimraf coverage build tmp`
`build`	Full production build	Includes memory optimization flag
`watch`	Development mode with file watching	`tsc -w -p tsconfig.json`
`pretest`	Pre-build quality checks	Runs fix, check, and compile
`start`	Build and run production	`npm run build && node build/src/index.js`
`run-local`	Run from build directory	Direct execution from build output
`build-local`	Quick local build and run	Compile and run without full build

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L6-L13)

package.json6-13 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L6-L13)

Testing Approach

Unit Testing Framework

The project uses Mocha for unit testing with TypeScript compilation:

```
npm run unittest  # Compiles and runs build/src/**/*_test.js
```

The testing workflow compiles TypeScript test files and executes them using `npx mocha` with glob pattern matching.

Testing Workflow:

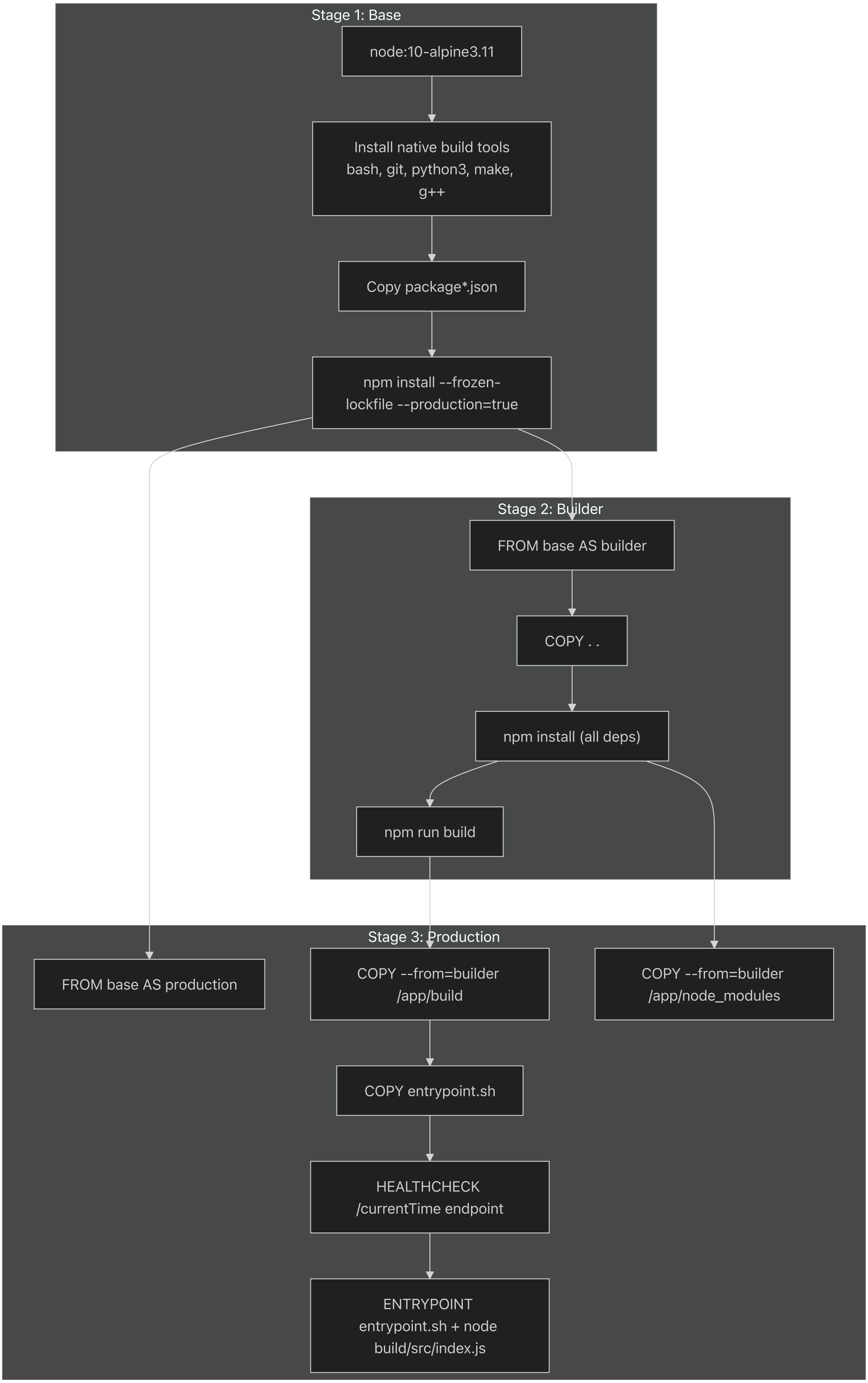
Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L18-L18) package.json18 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L18-L18)

Containerization Workflow

Multi-Stage Docker Build

The Docker build process uses a sophisticated multi-stage approach optimized for production deployment:

Docker Build Stages:


Stage 1: Base
node:10-alpine3.11
Install native build tools bash, git, python3, make, g++
Copy package*.json
npm install --frozen-lockfile --production=true
Stage 2: Builder
FROM base AS builder
COPY . .
npm install (all deps)
npm run build
Stage 3: Production
FROM base AS production
COPY --from=builder /app/build
COPY --from=builder /app/node_modules
COPY entrypoint.sh
HEALTHCHECK /currentTime endpoint
ENTRYPOINT entrypoint.sh + node build/src/index.js

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L56)

dockerfile1-56 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L1-L56)

Container Health Monitoring

The container includes a health check that validates the application's `/currentTime` endpoint every 30 seconds with a 3-second timeout and 3 retries before marking unhealthy.

```
node -e "require('http').get('http://localhost:3000/currentTime', (res) => {
```

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)

dockerfile53-54 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/dockerfile#L53-L54)

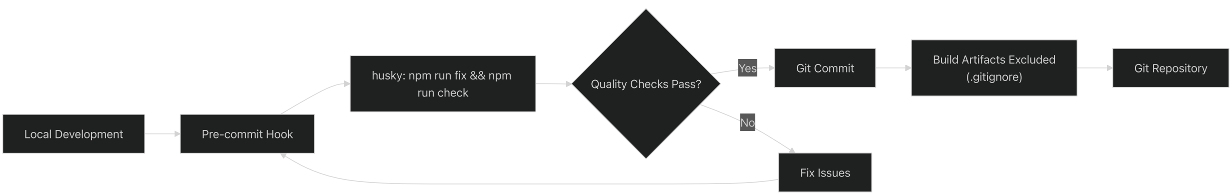
Version Control Practices

Git Ignore Configuration

The project excludes the following from version control:

Pattern	Purpose
`node_modules/**`	NPM dependencies
`.idea/**`	IntelliJ IDEA configuration
`.vscode/**`	Visual Studio Code settings
`build/**`	TypeScript compilation output
`src/env.js`	Environment-specific configuration
`logs/**`	Application log files

Version Control Workflow:



Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.gitignore#L1-L6)

.gitignore1-6 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/.gitignore#L1-L6)[

package.json56-60](https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60))

Pre-commit Quality Gates

The Husky configuration enforces code quality through pre-commit hooks that automatically run:

1. `npm run fix` - Auto-format code using GTS
2. `npm run check` - Validate code style compliance

This ensures all committed code meets project standards without manual intervention.

Sources: (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)

package.json56-60 (https://github.com/nhsv-tradex/lotte-ws-bridge/blob/08a647a0/package.json#L56-L60)
